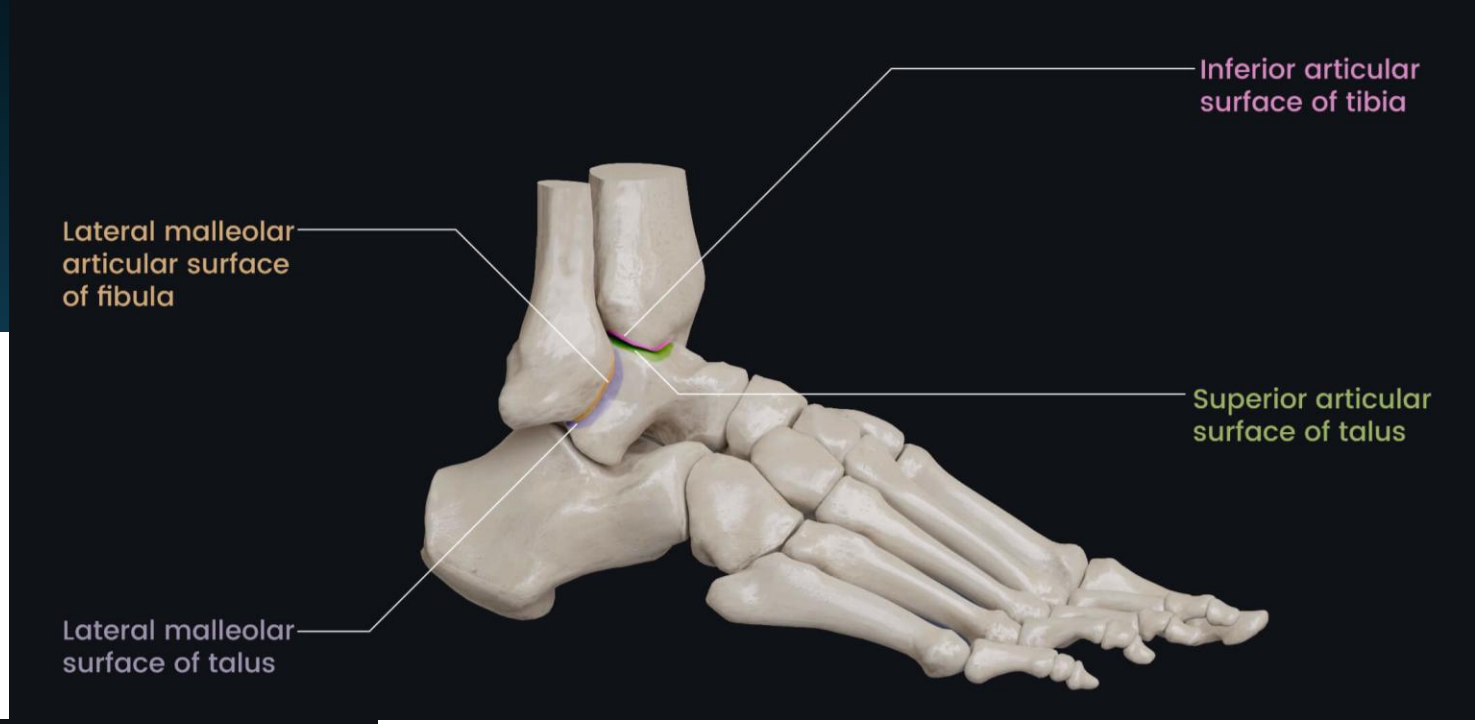
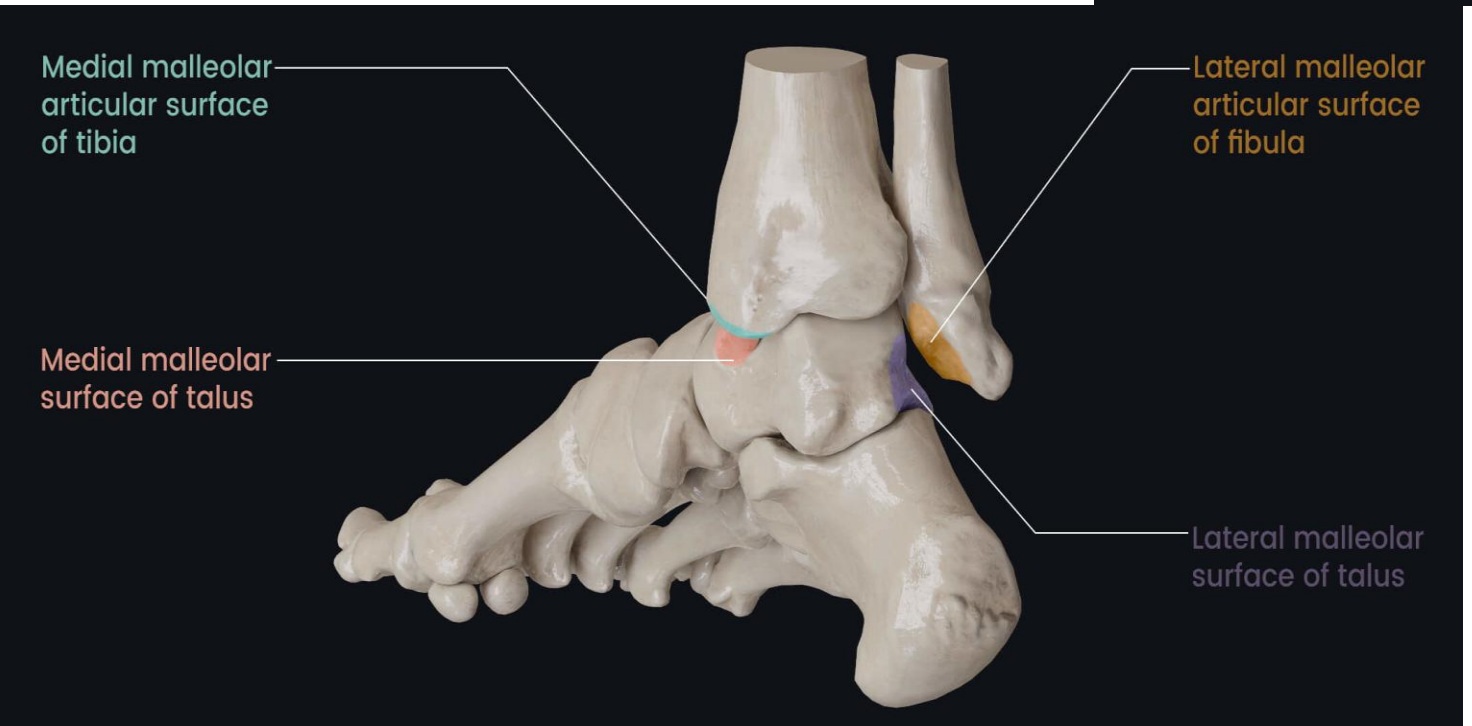
Anatomy of the Foot and Ankle

Walaa Abu-Taleb

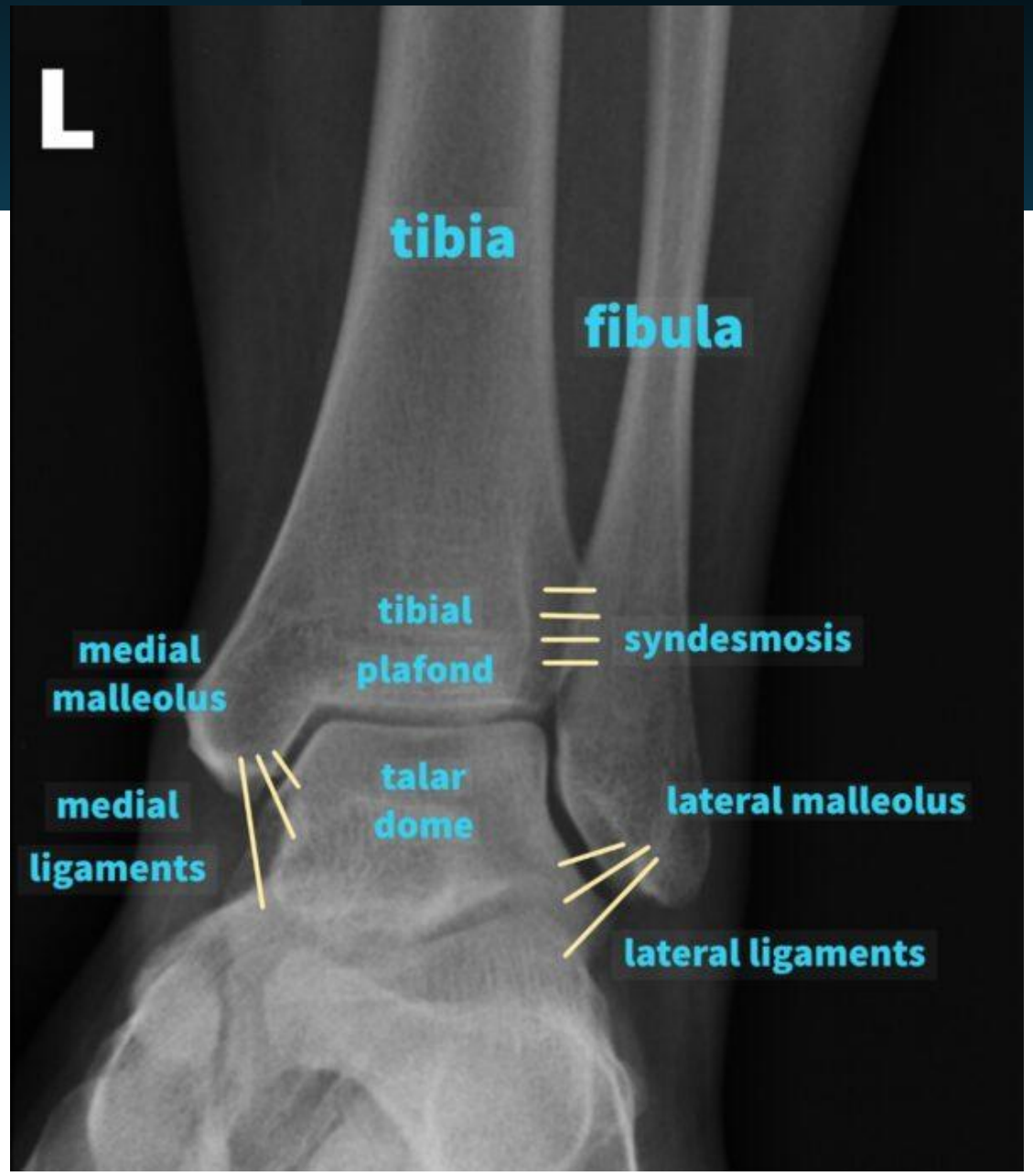


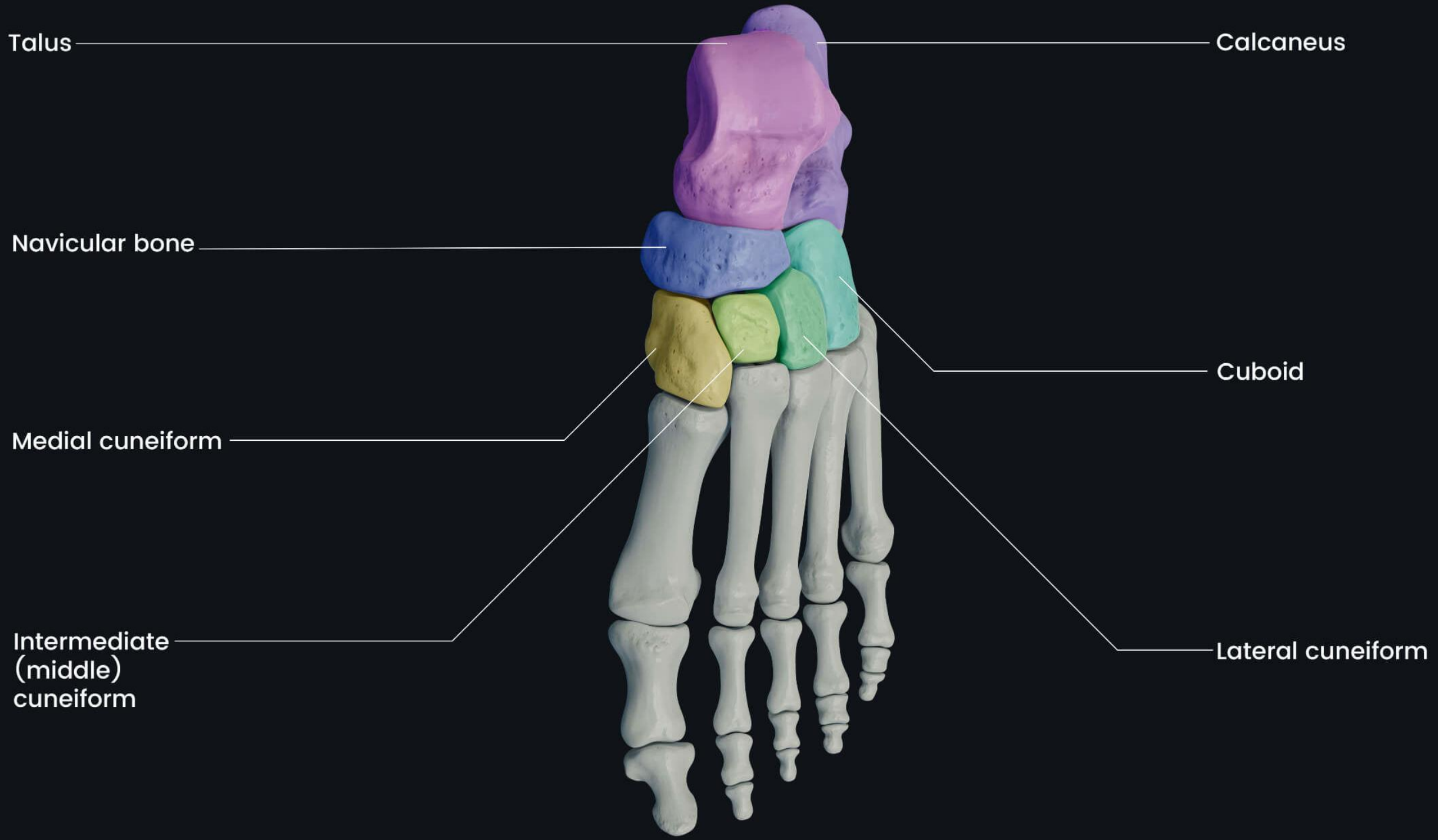
Bones of the Ankle





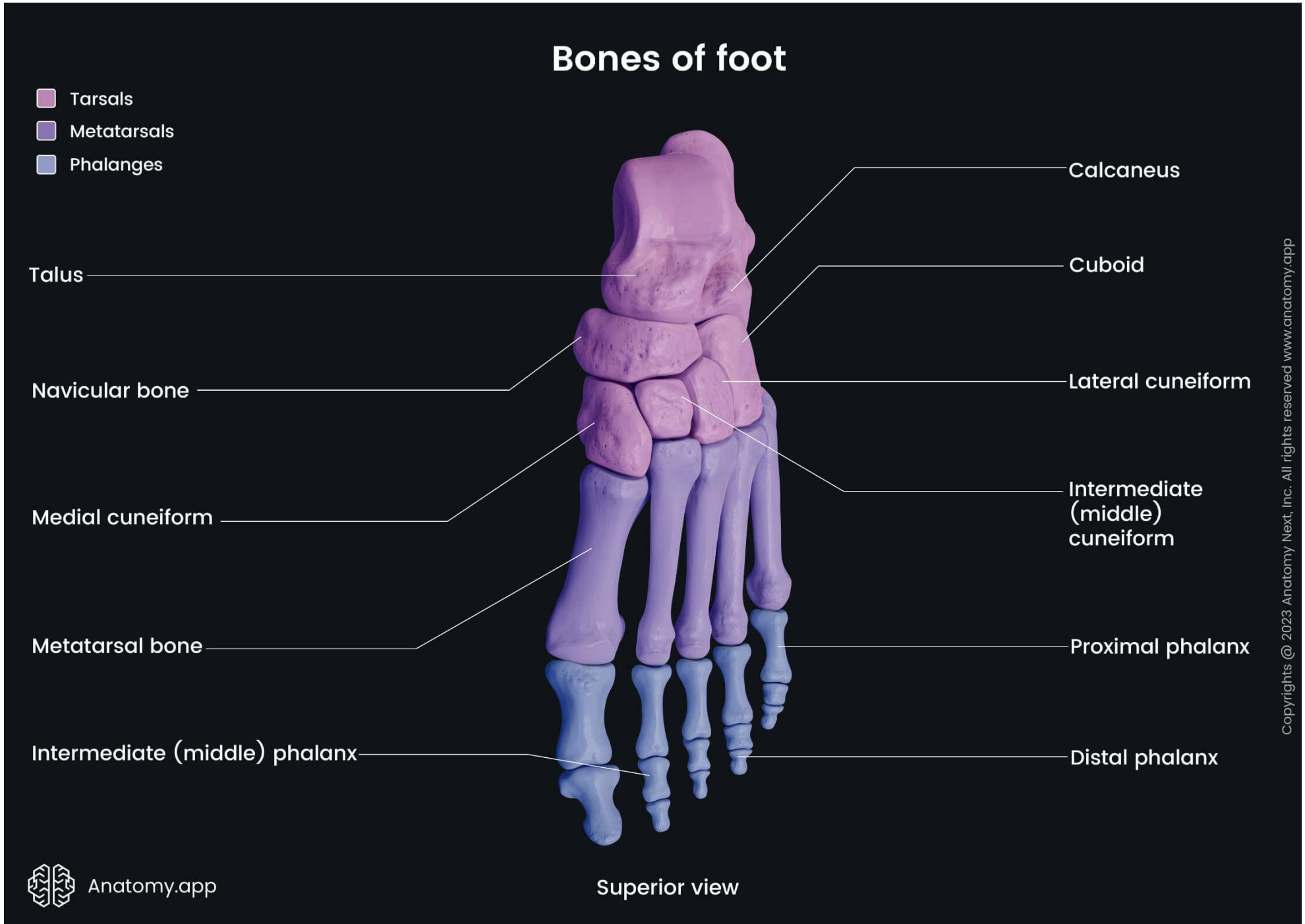
Bones of the Ankle



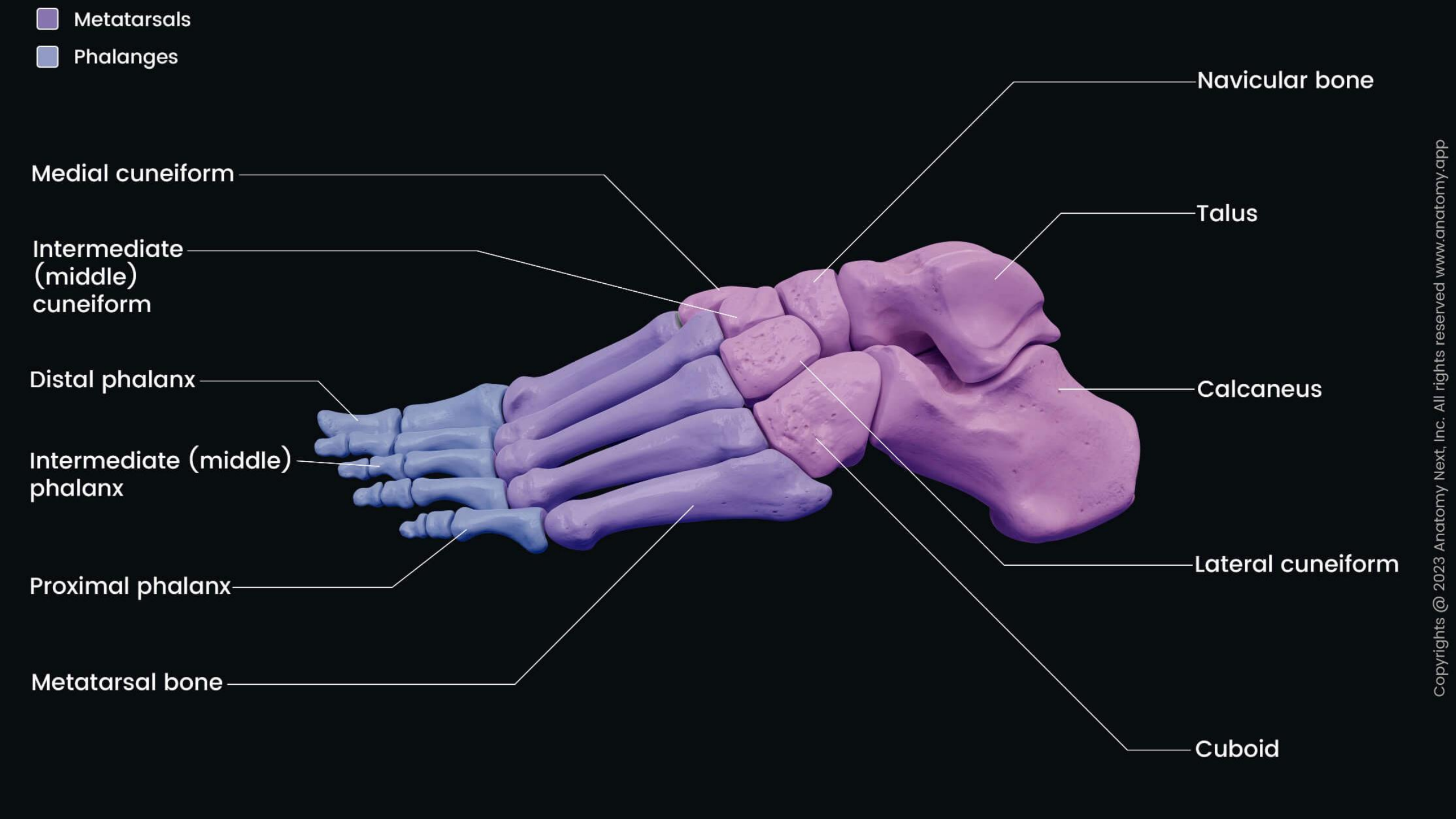


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Bones of foot



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Sesamoid Bones

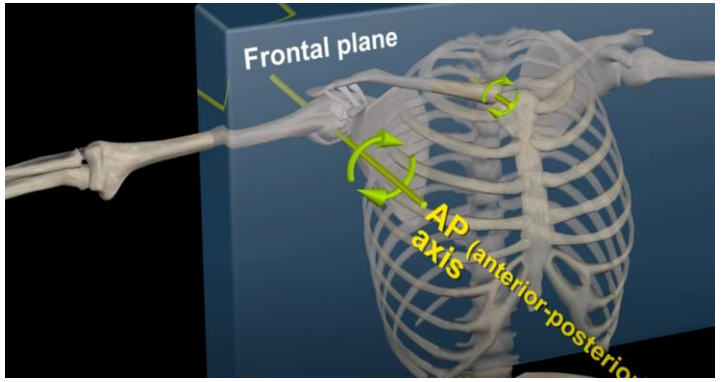
- A sesamoid bone is a small bone commonly found embedded within a muscle or tendon near joint surfaces, existing as focal areas of ossification and functioning as a pulley to alleviate stress on that particular muscle or tendon.
- The two semsamoid bones (medial and lateral) are two 1 x 1.5cm oval shaped bones that lie under the big toe joint, within the two tendons that move the big toe (flexor hallucis brevis).

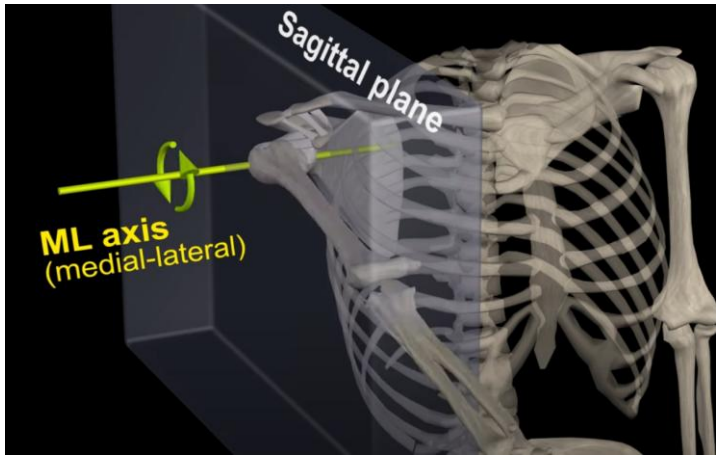


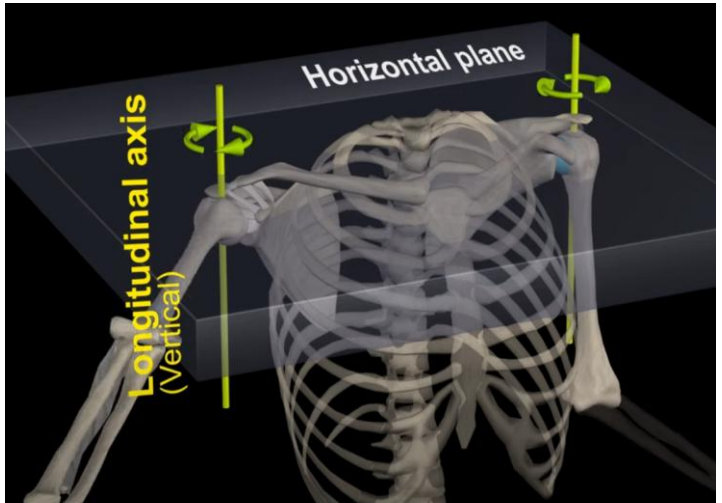
Axis of Rotation

The axis of rotation in a joint is an imaginary line that passes through the joint's pivot point. The motion occur at a perpendicular angle to axis of rotation. There are three axes of rotation in the human body:

- **Anterior-posterior (sagittal) axis**: this axis allows for movement in the frontal plane.
- **Mediolateral (transverse) axis**: this axis allows for movement in the sagittal plane.
- **Longitudinal (Vertical) axis**: This axis allows for rotation in the transverse plane.

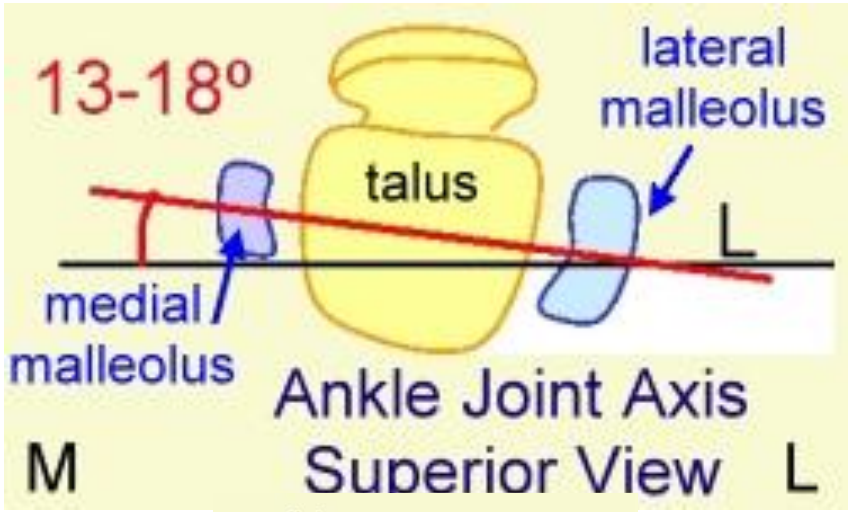


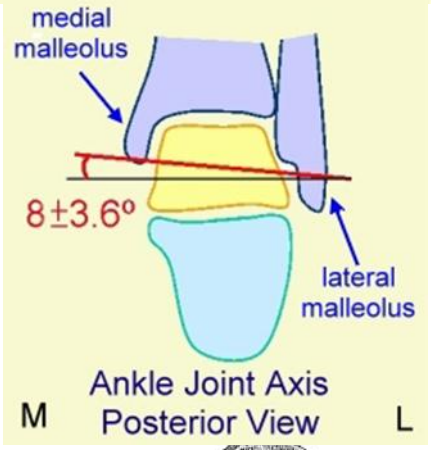


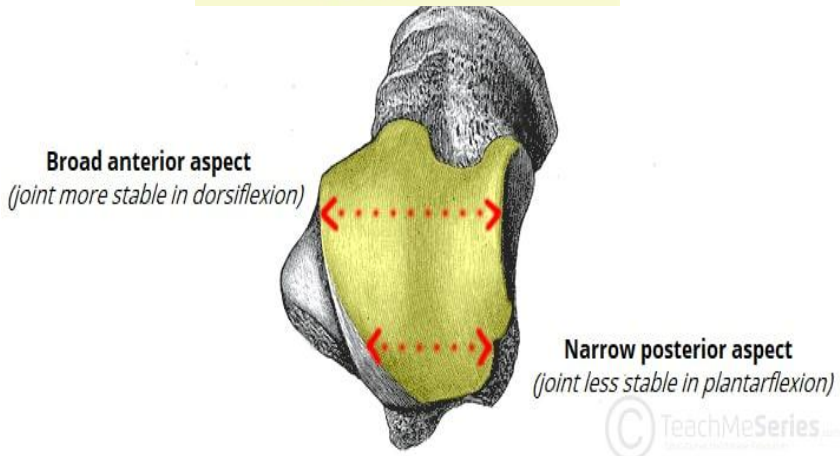


Talocrural Joint

- The tip of the medial malleoli is anterior and superior to the lateral malleoli, which makes its axis oblique to both the sagittal and frontal planes.
- Motion in other planes is required (like horizontal and frontal plane) to achieve a complete motion for plantarflexion and dorsiflexion.
- The reported normal available range for dorsiflexion varies in the literature between 0°-16.5°and 0°-25°, and this changes with weightbearing.
- The normal range of plantarflexion has been reported to be around 0°-50°.
- The talus rolls within the mortise during dorsiflexion and plantarflexion. During dorsiflexion, the talus rolls anteriorly and it glides posteriorly. While with plantarflexion, the talus rolls posteriorly and glides anteriorly.

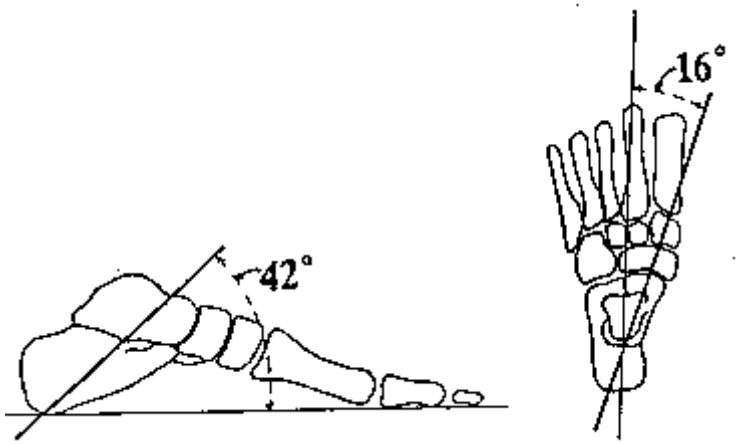








Subtalar Joint



- Most dynamic musculoskeletal models define the subtalar joint as a one degree of freedom hinge with a tri-planar axis.
- The axis of the subtalar joint lies about 42° superiorly to the sagittal plane and about 16° to 23° medial to the transverse plane.
- Ratio of inversion-to-eversion movement is about 2:1 and a 3:2.
- Secondary to the anatomy of the subtalar joint, the coupled motion of dorsiflexion, abduction and eversion produces pronation, whereas the coupled motion of plantarflexion, adduction and inversion produces supination.
- Pronation is how your body absorbs the forces that result from your foot hitting the ground. Supination is how it translates those forces into forward motion.
- The average ROM for pronation is 5° and 20° for supination. Inversion and eversion ROM has been identified as 30° and 18°, respectively.

Lower Leg Muscle Compartments

The lower leg muscles are divided into four compartments:

1. The superficial posterior compartment,
2. The deep posterior compartment,
3. The lateral compartment,
4. And the anterior compartment.

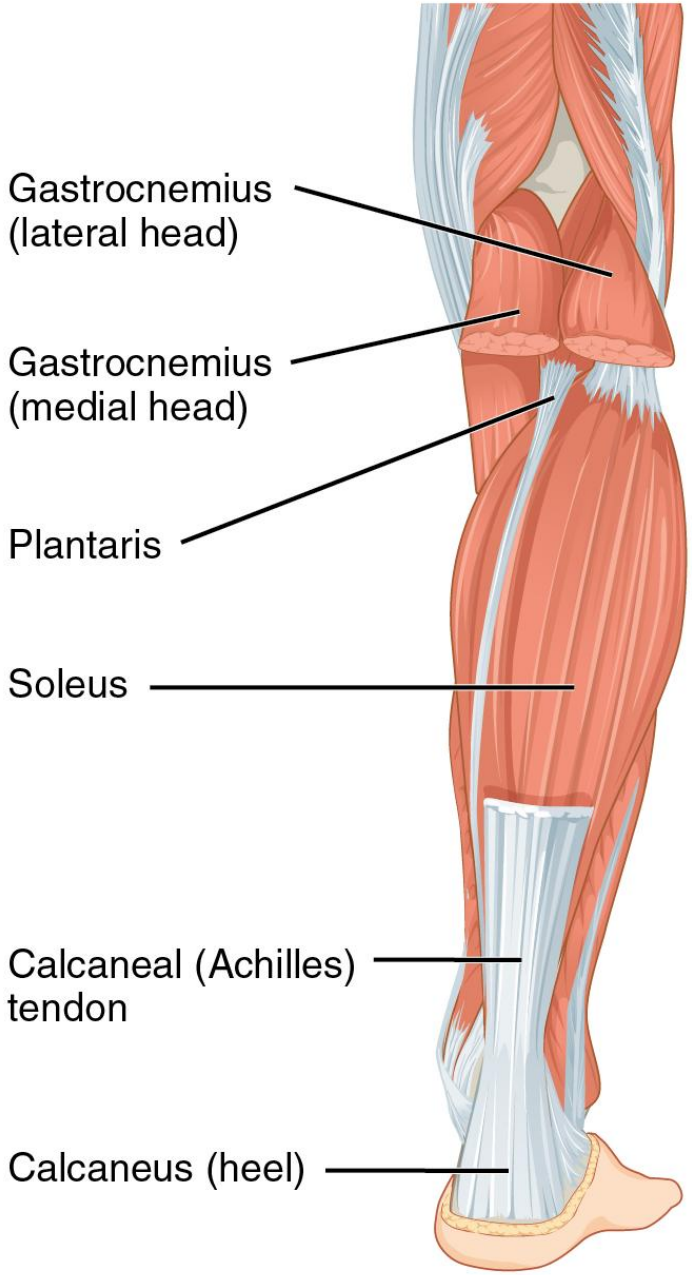
Posterior Compartments (The primary plantar flexors)

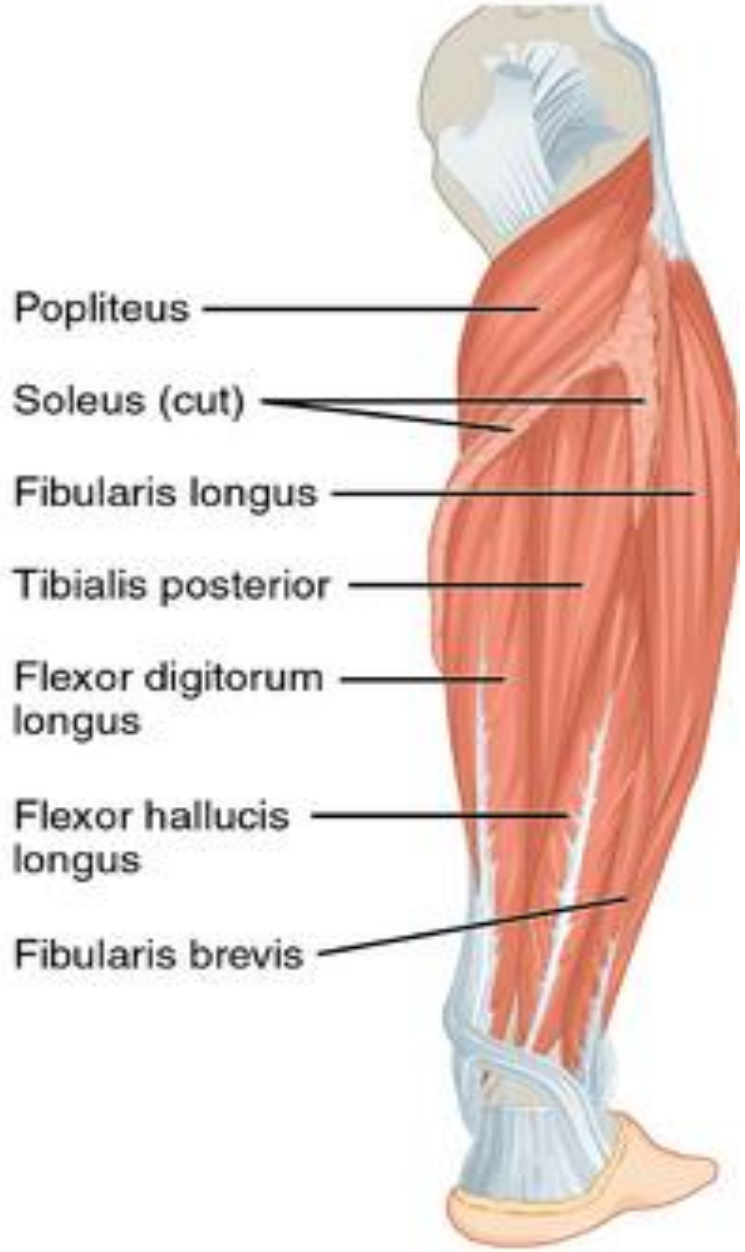
Primary responsibilities include:

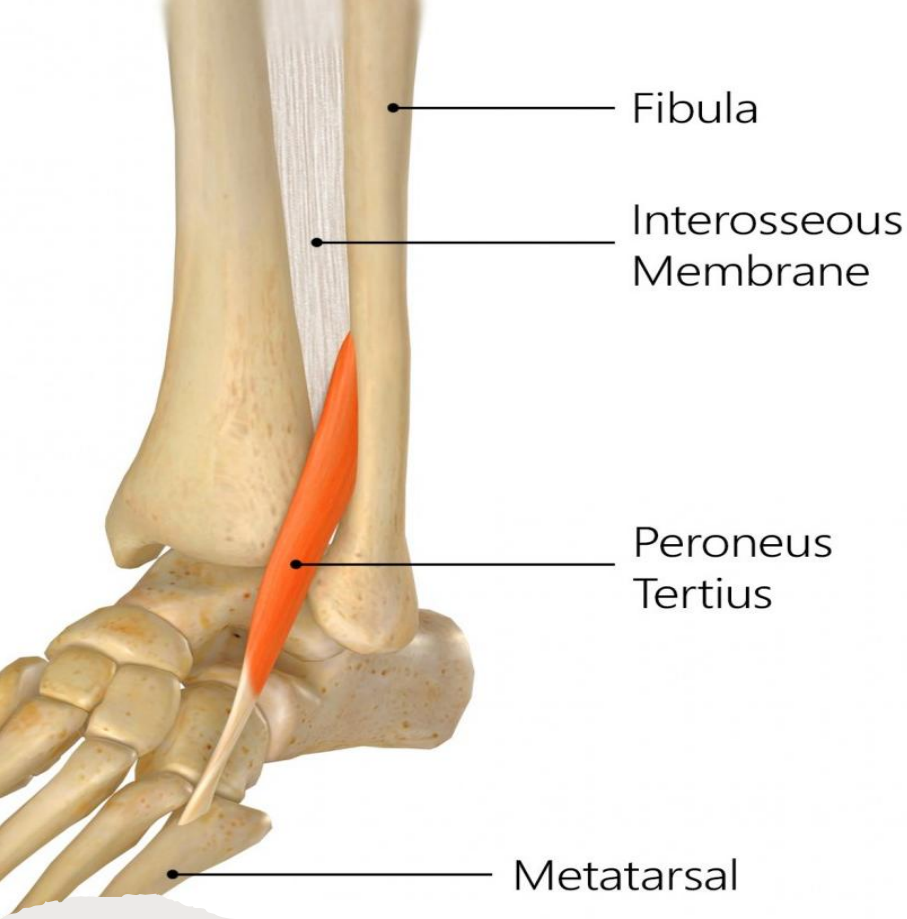
- Transforming the foot into a rigid lever
- Assisting with push-off during the gait cycle
- Controlling tibia progression over the foot during initial contact through the push-off gait cycle
- Controlling foot pronation during initial contact through the push-off gait cycle

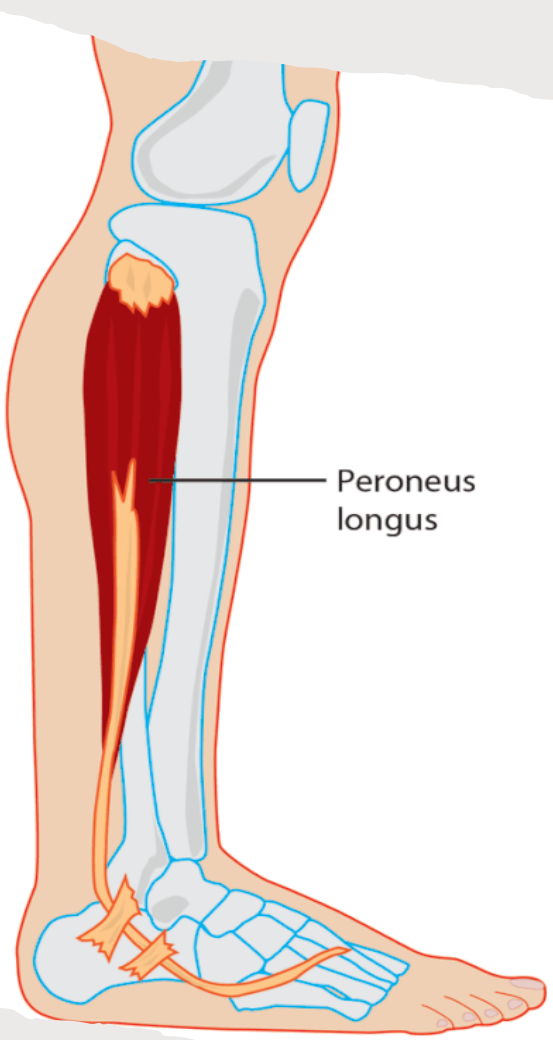


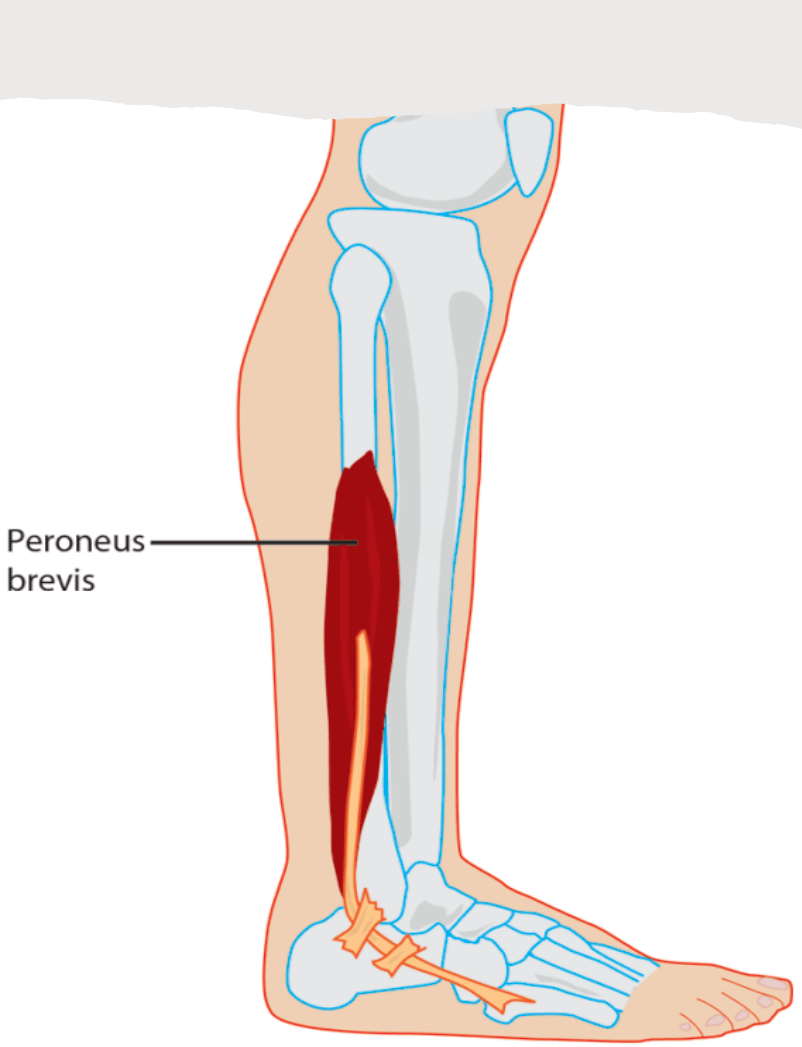
Posterior Compartment







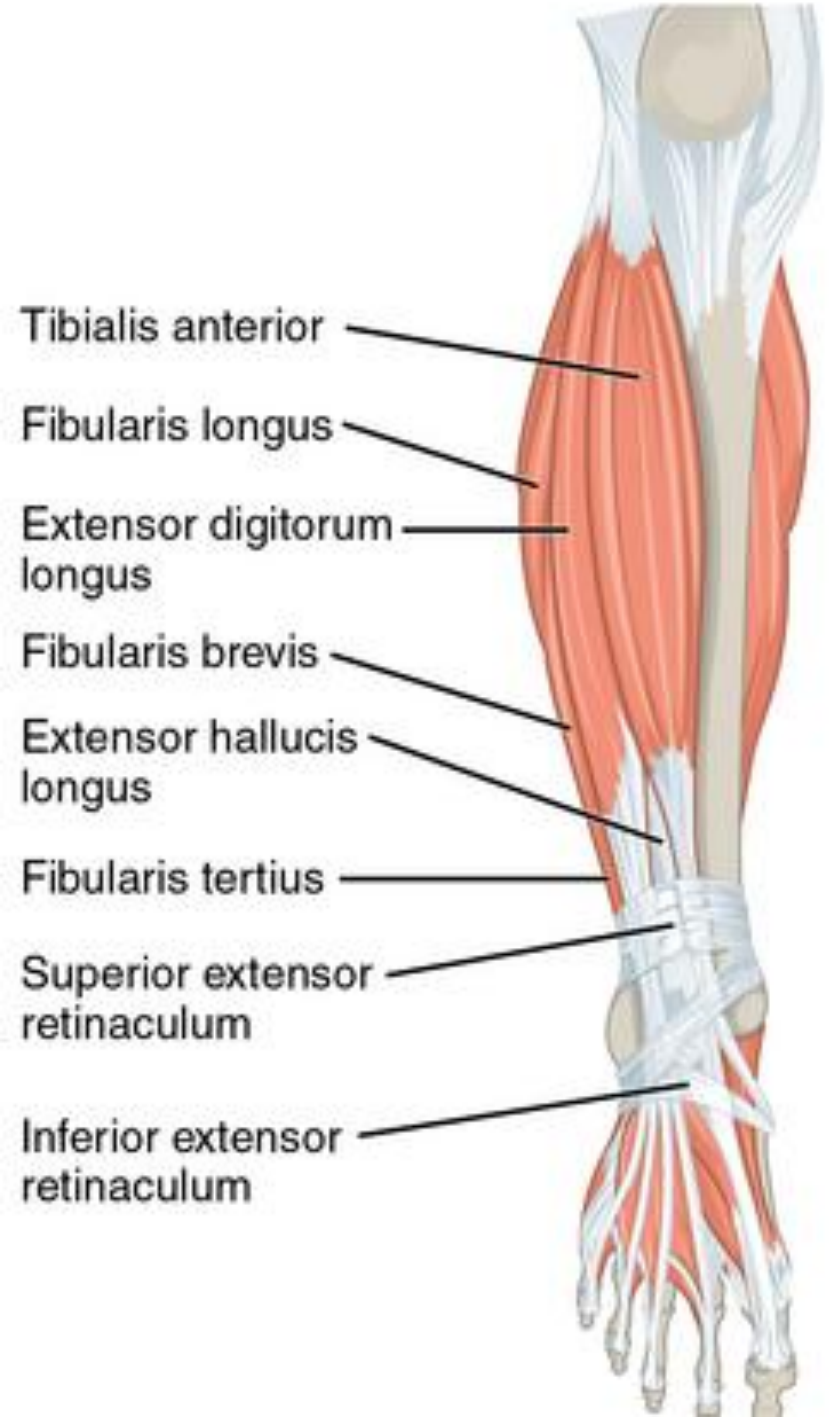




Lateral Compartment

Anterior Compartment

- Tibialis anterior and extensor hallucis longus invert the foot during dorsiflexion.
- Extensor digitorum longus everts the foot during dorsiflexion.
- Eccentric control of foot lowering during heel strike.
- Concentric control of toes clearance during swing phase.



Anterior compartment of the leg



Activity

How to perform a grade 4 Muscle strength Testing for:

1- Flexor hallucis longus?

2- Medial head of gastrocnemius?

3- Extensor digitorum Longus

4- Fibularis tertius?

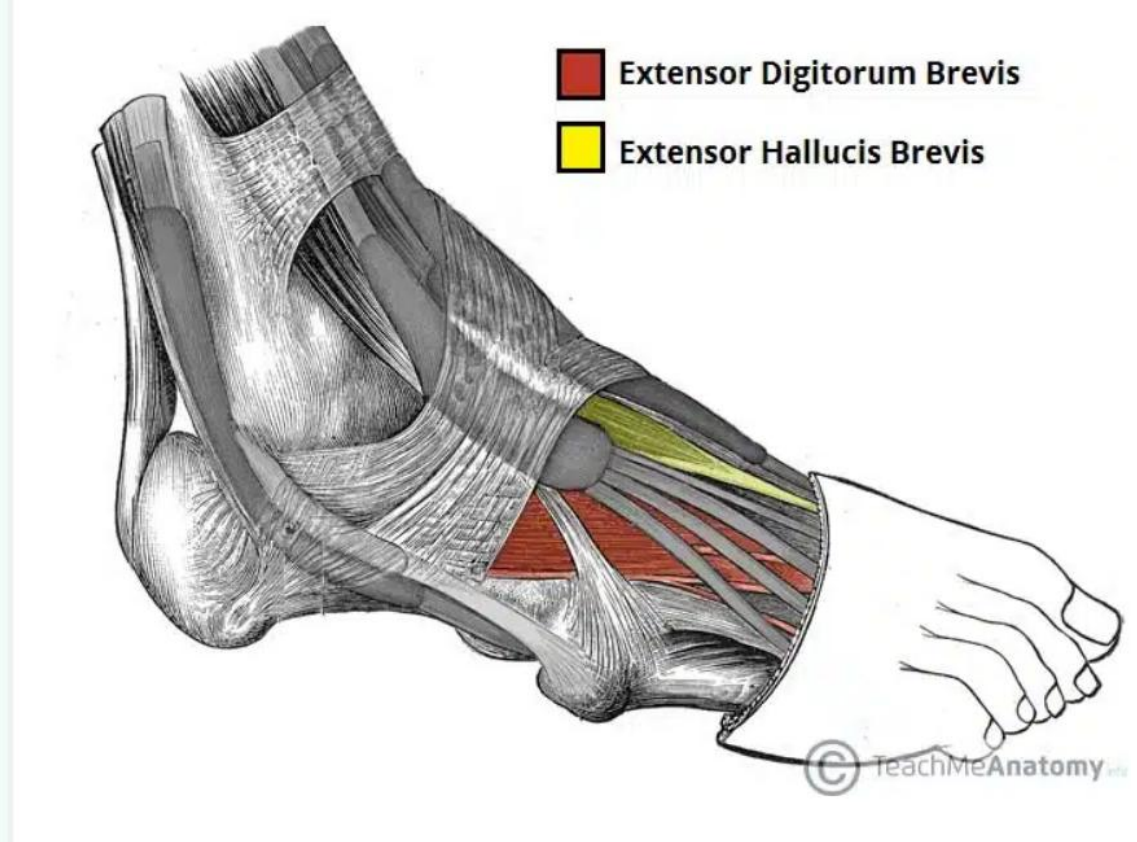
The foot muscles can be divided into

- **Extrinsic** muscles arise from the anterior, posterior and l ateral compartments of the leg. They are mainly responsible for actions such as eversion, inversion, plantarflexion and dorsiflexion of the foot.
- **Intrinsic** muscles are located within the foot and are responsible for the fine motor actions of the foot, for example movement of individual digits.

Extrinsic Muscles

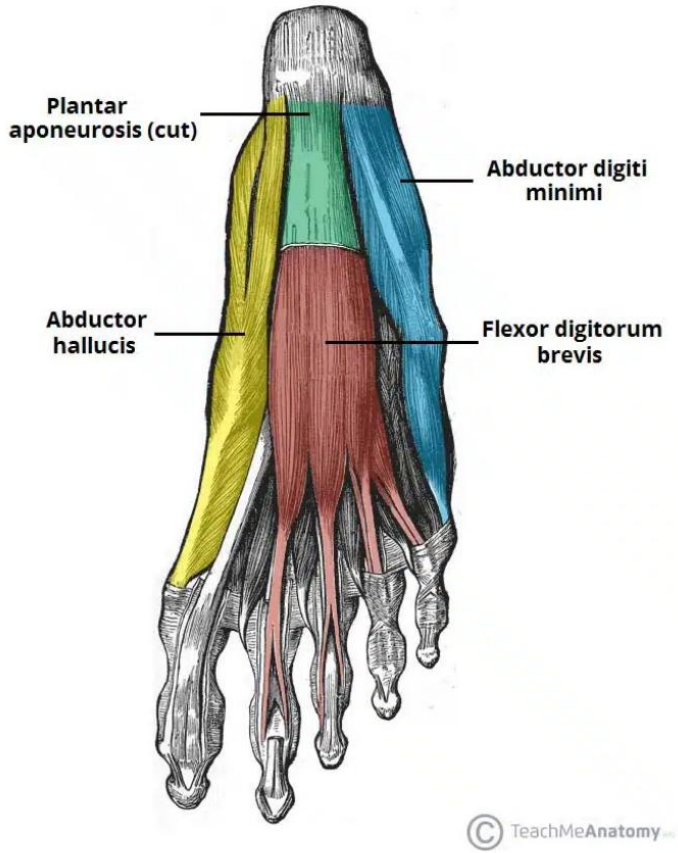
Dorsum of the foot

- **Extensor Digitorum Brevis**
- **Extensor Hallucis Brevis**



Intrinsic Muscles (Planter aspect)

- First Layer:
- 1- Abductor Hallucis
- 2- Flexor Digitorum Brevis
- 3- Abductor Digiti Minimi



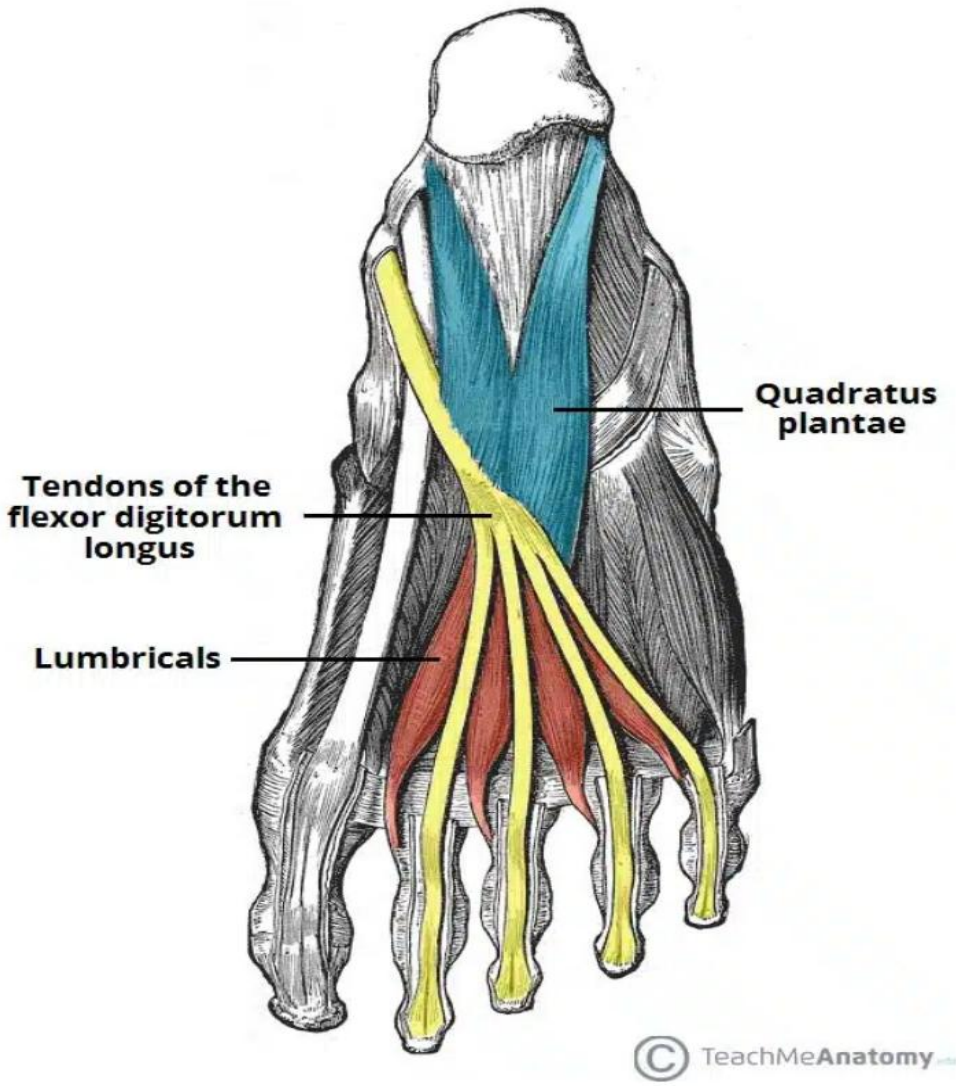
Intrinsic Muscles (Planter aspect)

Second Layer:

1- Quadratus Plantae

2- Lumbricals

- quadratus plantae helps with toe flexion in the sagittal plane
- lumbricals extend the interphalangeal joint



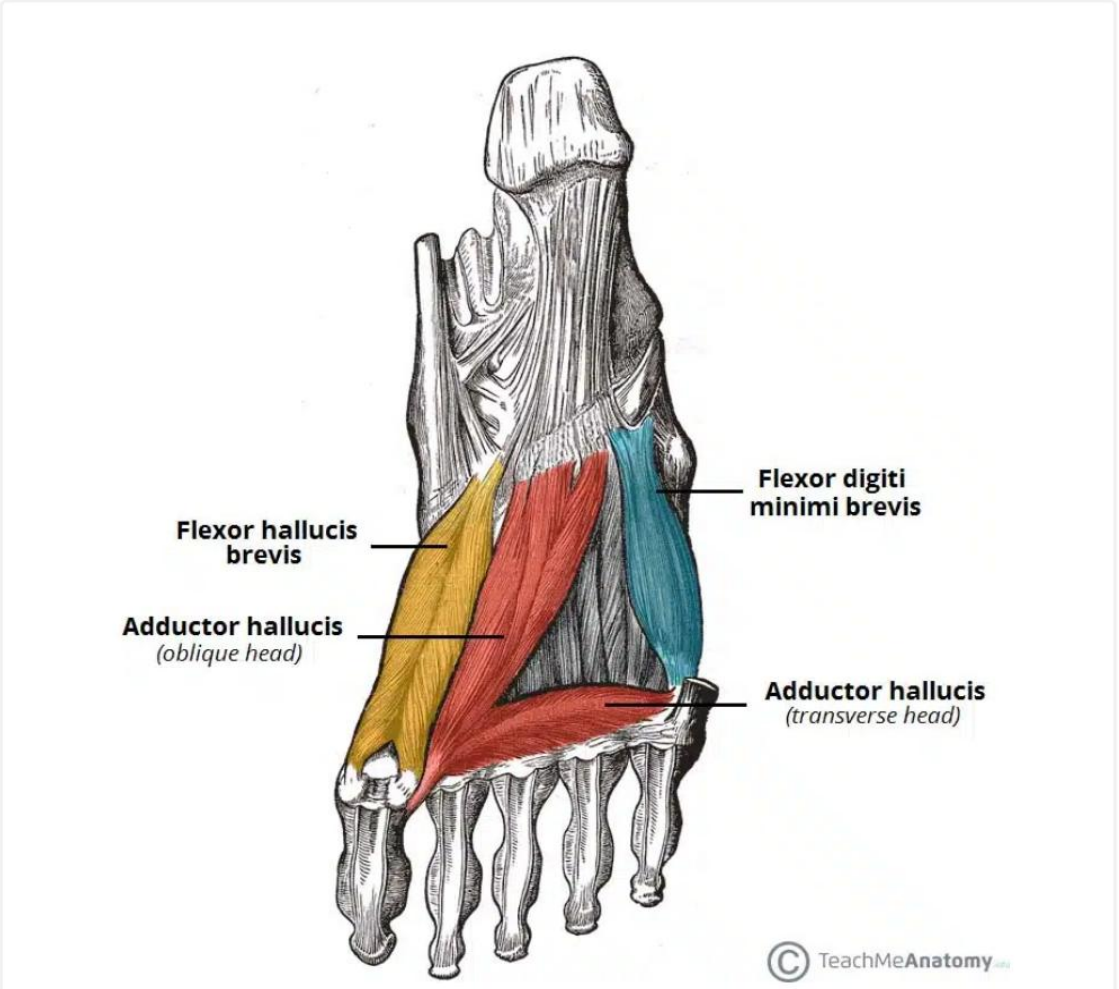
Intrinsic Muscles (Planter aspect)

Third Layer:

1- Flexor Hallucis Brevis

2- Adductor Hallucis

3- Flexor Digiti Minimi Brevis



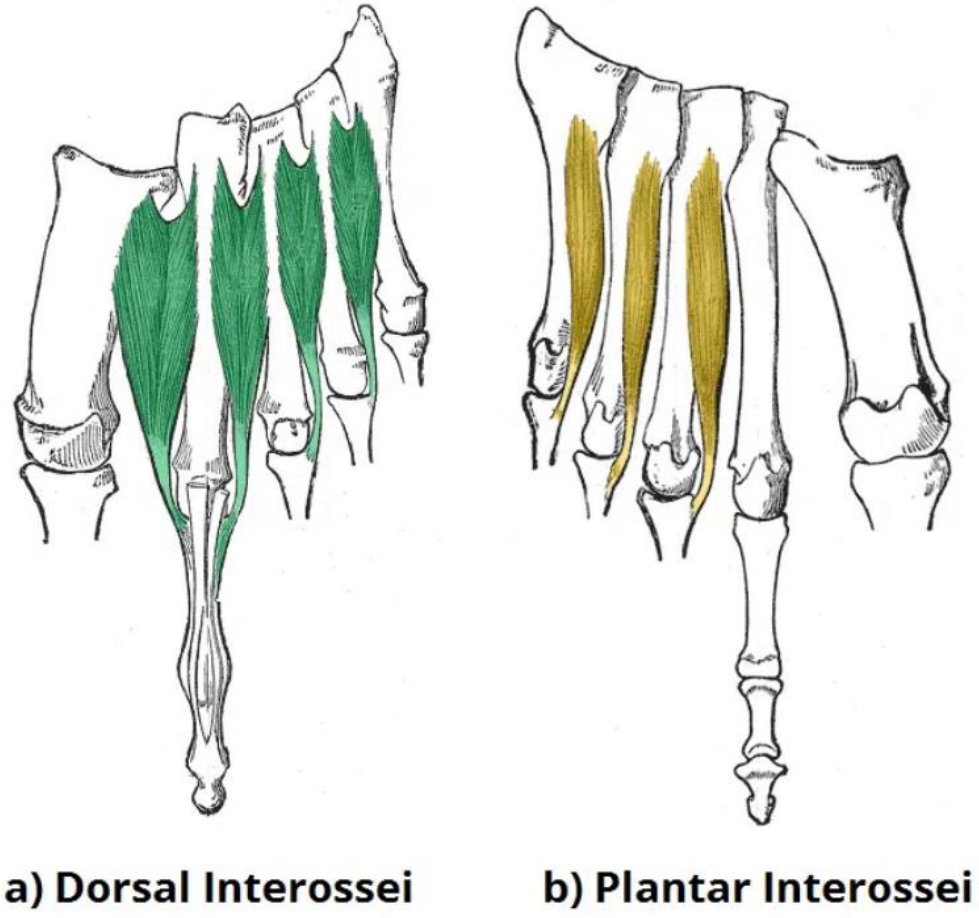
Intrinsic Muscles (Planter aspect)

Fourth Layer:

1- Plantar Interossei

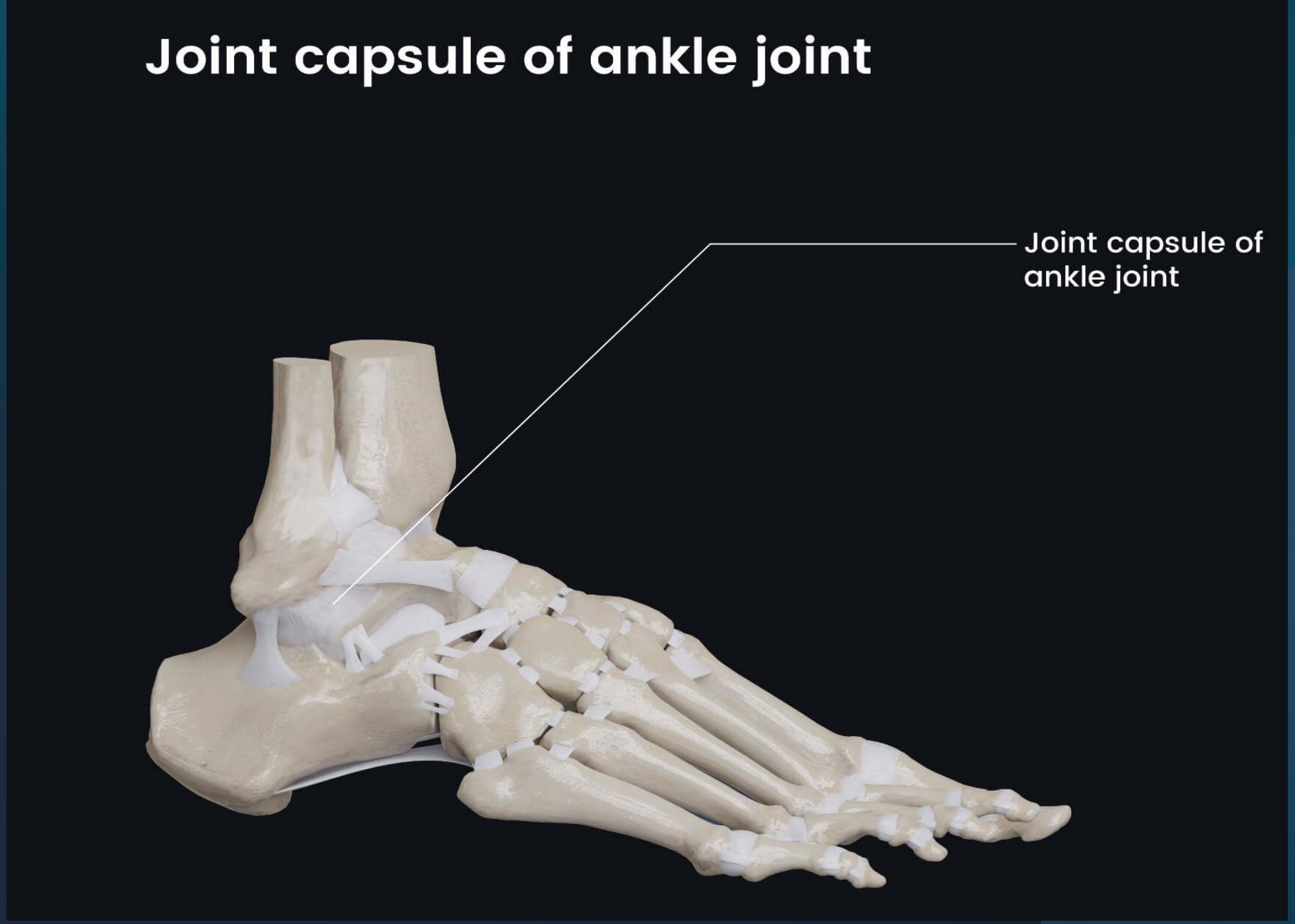
2- Dorsal Interossei

- interosseus assist with the MTP joint flexion, IP joint extension, toe abduction and adduction

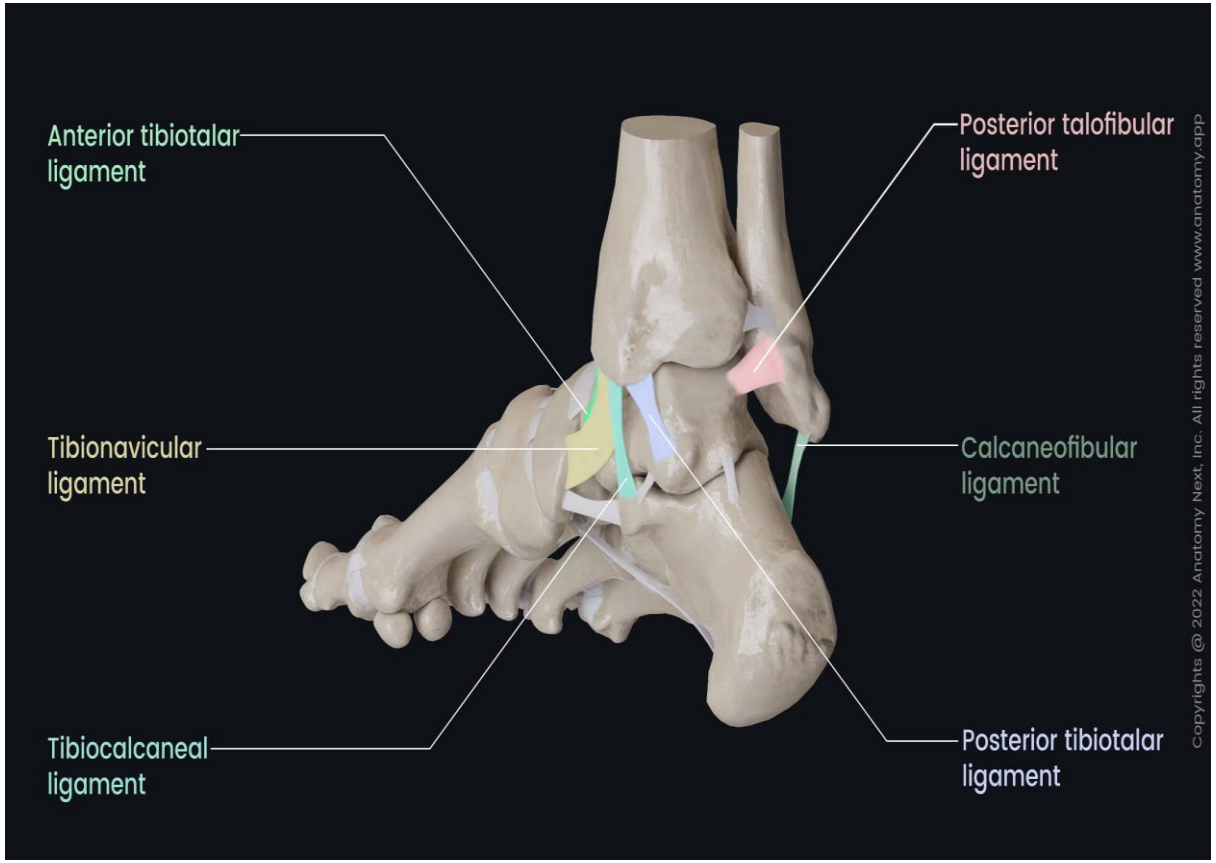


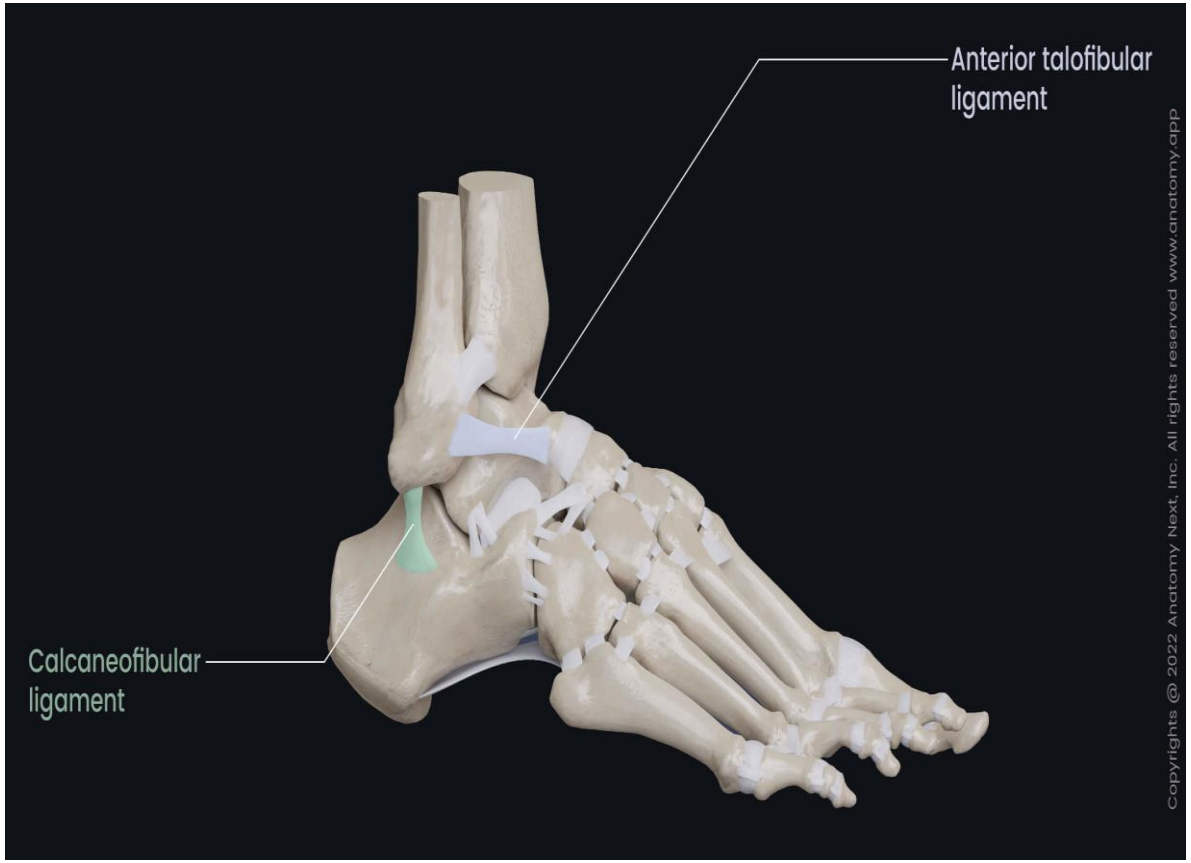
a) Dorsal Interossei b) Plantar Interossei

Joint capsule of ankle joint



Ligaments of the Ankle joint





Ankle Instability

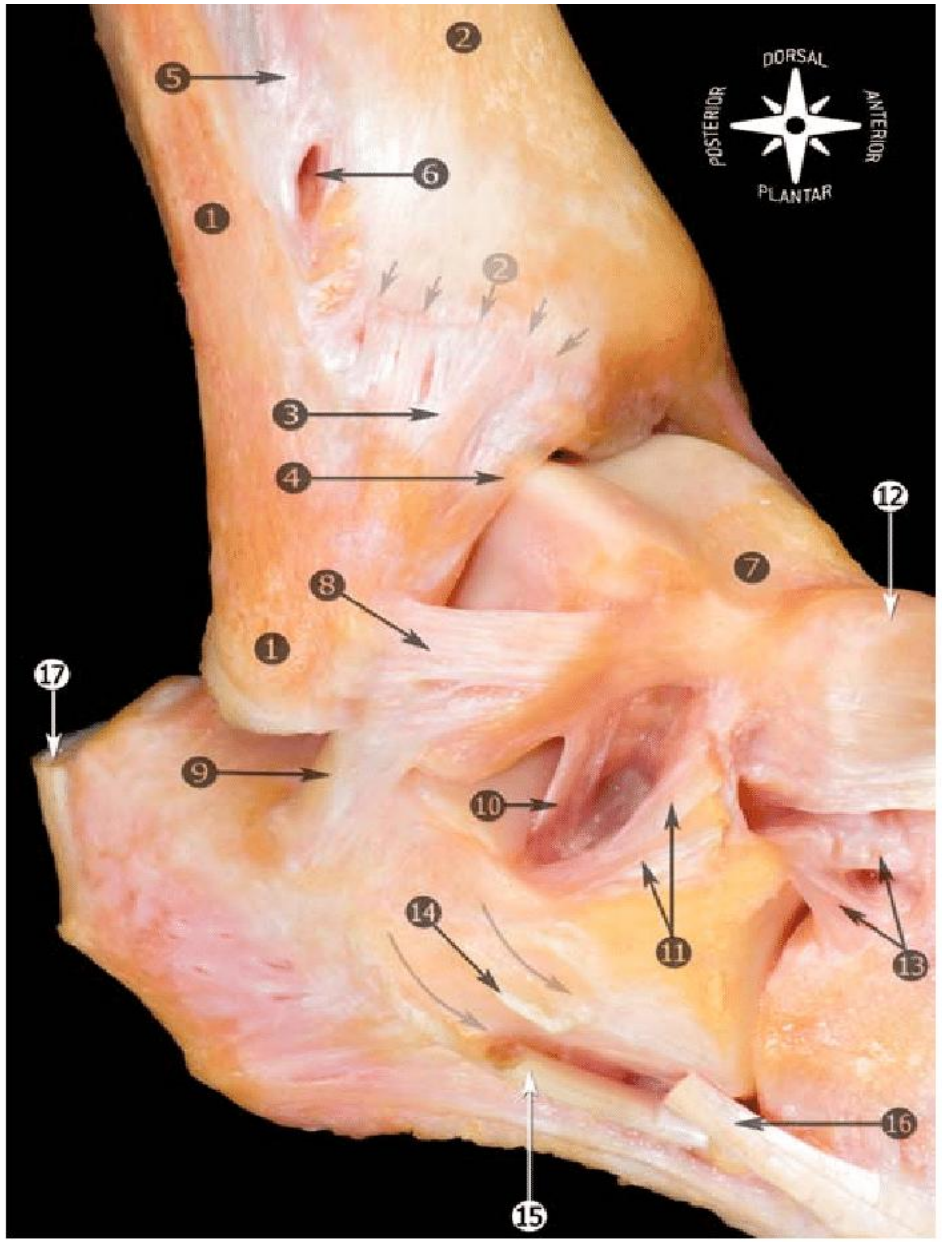




Ankle Instability

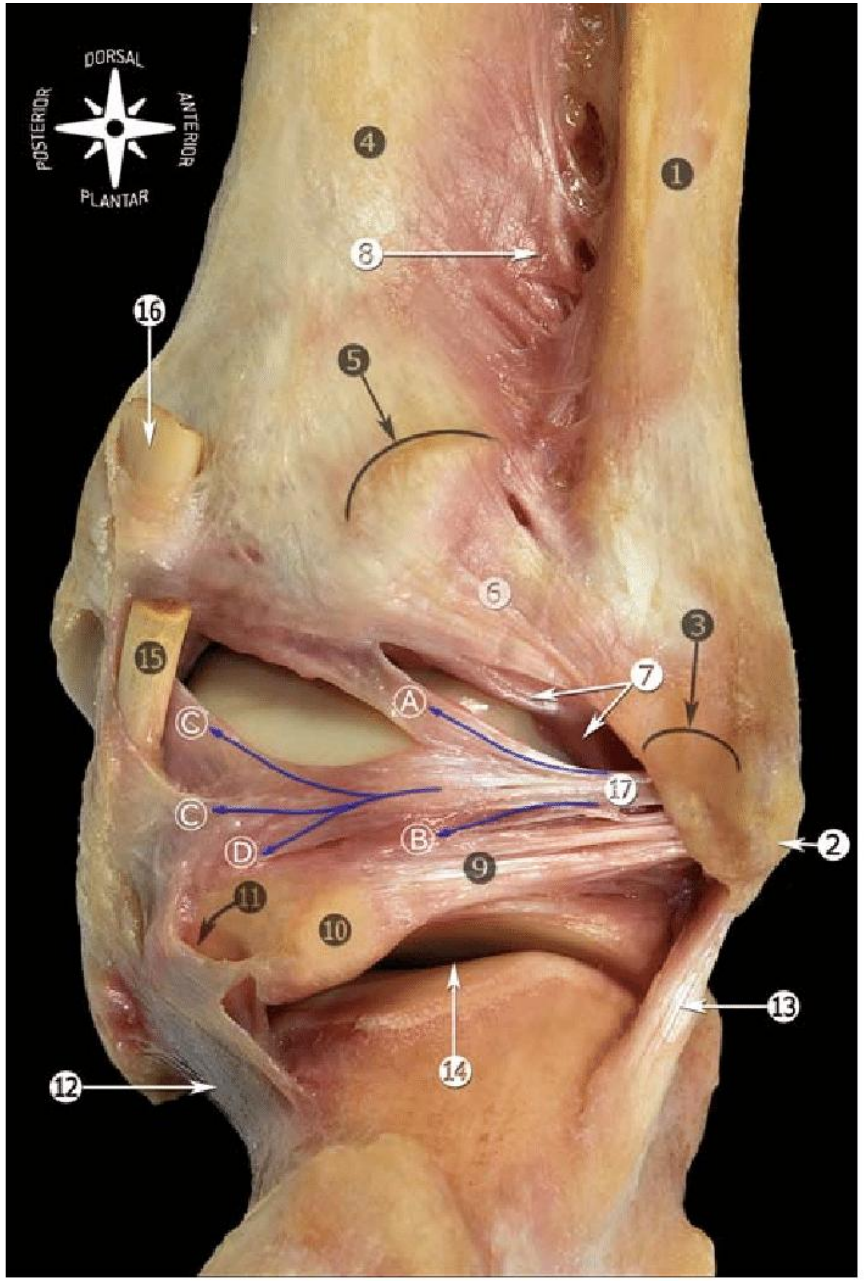






Anatomic dissection of the lateral region of the foot and ankle showing the morphology and relationship of the anterior talofibular with the calcaneofibular ligaments.

1 Fibula and tip of the fibula;
2 tibia (anterior tubercle with arrows);
3 anterior tibiofibular ligament;
4 distal fascicle of the tibiofibular ligament;
5 interosseous membrane;
6 foramen for the perforating branch of the peroneal artery;
7 talus;
8 anterior talofibular ligament;
9 calcaneofibular ligament;
10 talocalcaneal interosseous ligament;
11 inferior extensor retinaculum (cut);
12 talonavicular ligament;
13 bifurcate ligament;
14 peroneal tubercle (arrows showing the peroneal tendons sulcus);
15 peroneus longus tendon;
16 peroneus brevis tendon;
17 calcaneal tendon



Posterior view of the anatomic dissection of the ankle ligaments showing the posterior intermalleolar ligament with its relation to the surrounding anatomy.

- 1 Fibula;
- 2 tip of the fibula;
- 3 peroneal groove of the fibula;
- 4 tibia;
- 5 posterior tubercle of the tibia;
- 6 superficial component of the posterior tibiofibular ligament;
- 7 deep component of the posterior tibiofibular ligament or transverse ligament;
- 8 interosseous membrane;
- 9 posterior talofibular ligament;
- 10 lateral talar process;
- 11 tunnel for flexor hallucis longus tendon;
- 12 flexor hallucis longus retinaculum;
- 13 calcaneofibular ligament;
- 14 subtalar joint;
- 15 flexor digitorum longus tendon (cut);
- 16 tibialis posterior tendon (cut);

17 posterior intermalleolar ligament: A Tibial insertion (tibial slip in arthroscopic view). B Talar insertion (lateral talar process). C Tibial malleolar insertion through the septum between the flexor digitorum longus and posterior tibial tendons. D Talar insertion (medial talar process) through the joint capsule

Clinical insights of the ankle and foot

Activity

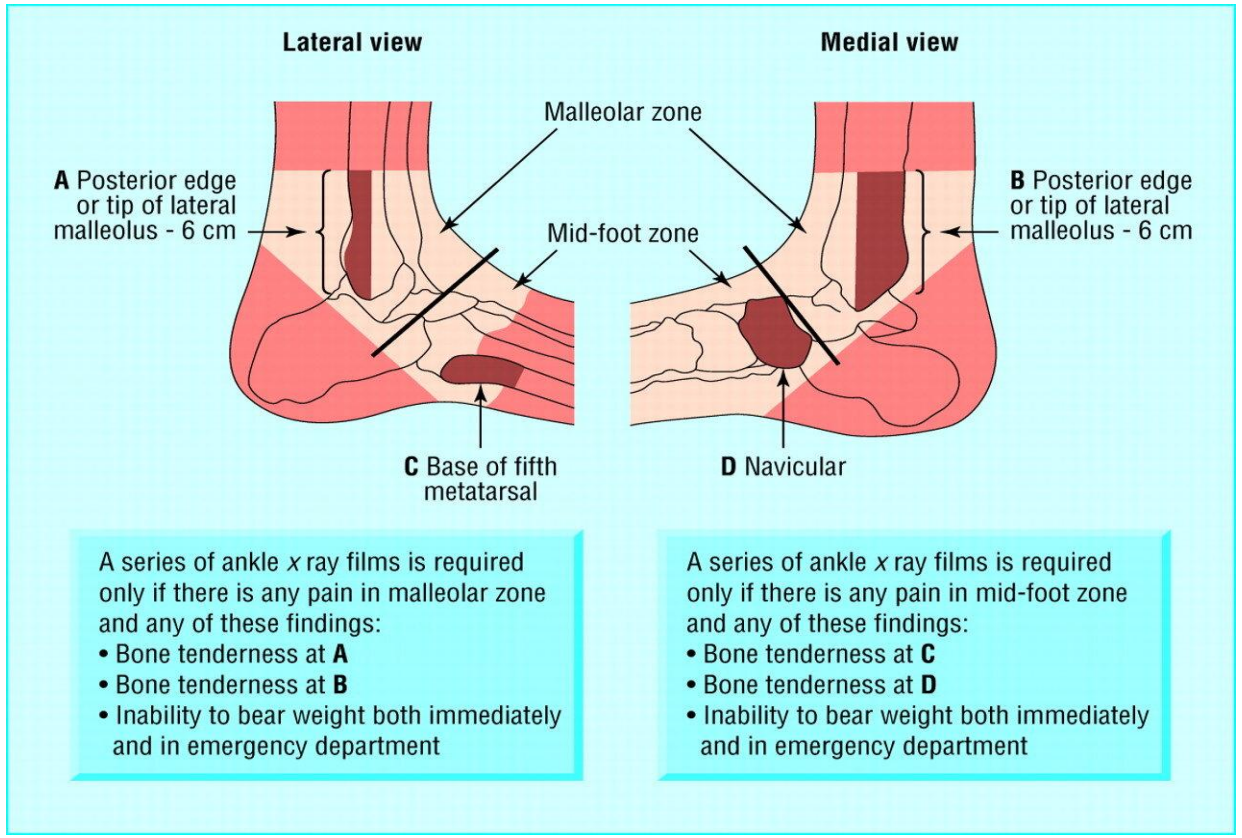
1- What are the most common fracture sites in the ankle and foot?

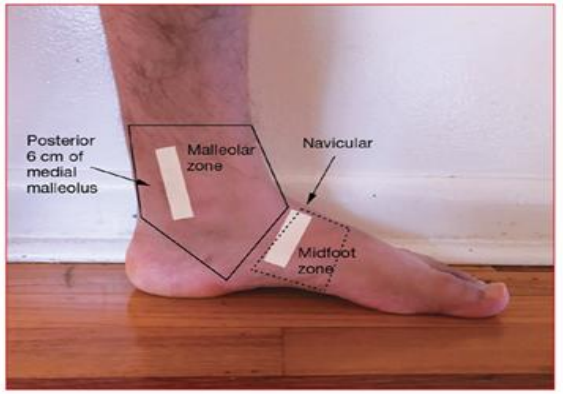
Hint:

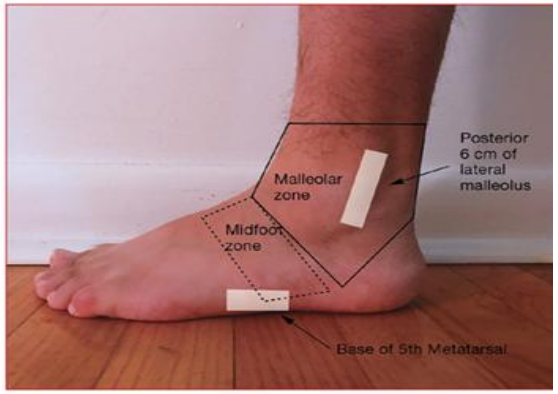
2- How to use palpation to clinically diagnose fracture of the ankle and foot?

Ankle and Foot Fracture (#)

- Caused by falls, twisting, or direct impact of a foot against a hard object, cause considerable pain, which is usually made worse by putting weight on the foot.
- Needs x-rays to diagnose foot # except for certain toe #. Rarely, computed tomography (CT) or magnetic resonance imaging (MRI) is required.
- Treatment depends on the bone and the type of fracture (#)
- Treatment involves a splint or a shoe or boot specially designed to protect the foot.





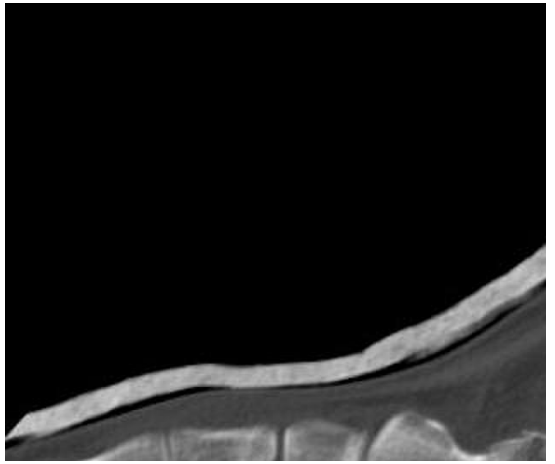


Fracture of the foot

- # of Calcaneus ?
- # of Talus?
- # of Navicular?



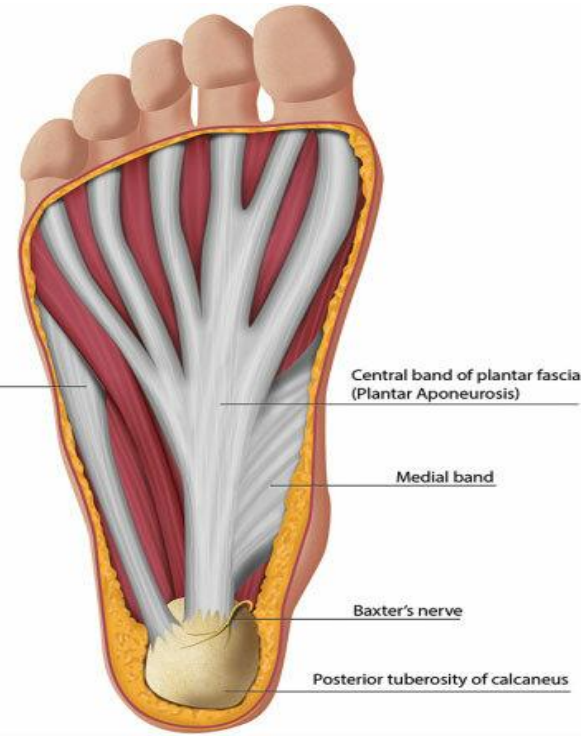


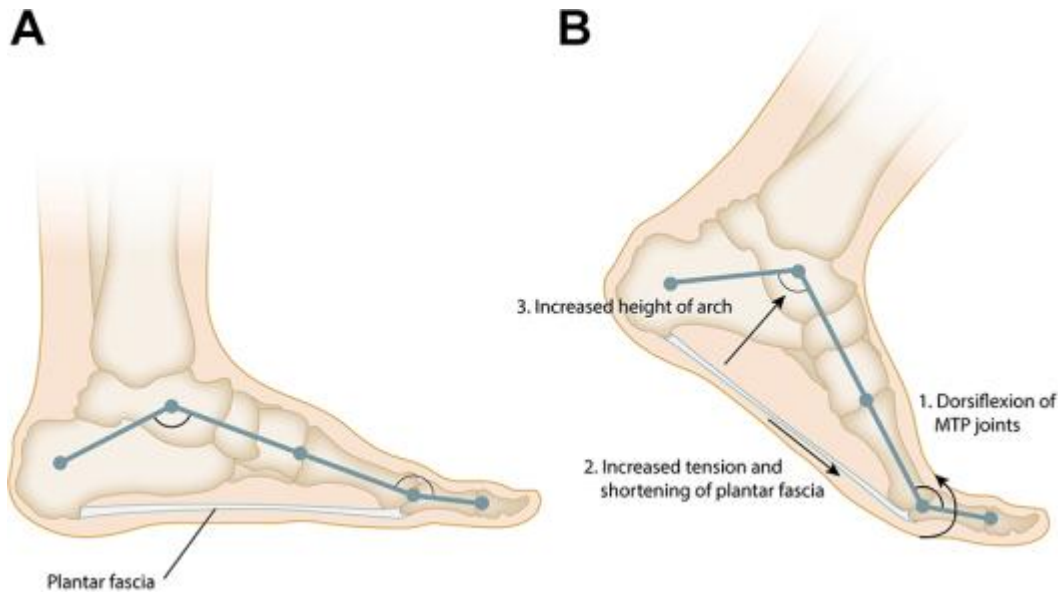


Plantar fascia

- The plantar fascia originates at the medial tubercle of the calcaneus and inserts at 3 locations in the forefoot, creating 3 distinct bands: medial, central, and lateral.
- The medial band overlies and inserts onto the muscles of the hallux, and the lateral band inserts on the base the fifth metatarsal.
- The central band (aka the plantar aponeurosis) is the thickest, strongest, and most often involved in Planter fasciitis.
- The plantar fascia is responsible for raising and stabilizing the arch during gait via the windlass mechanism.

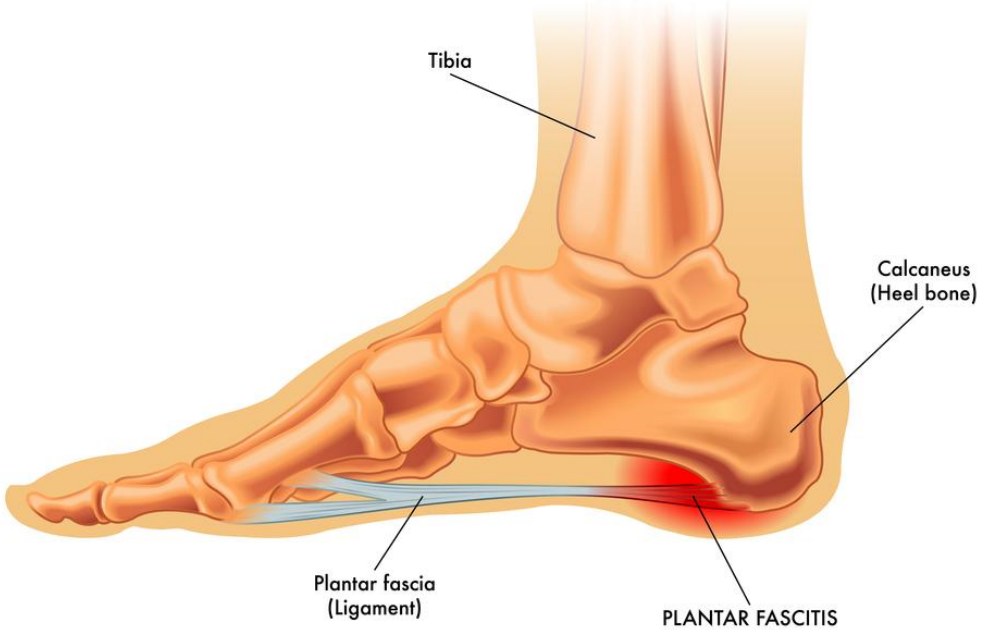






Plantar Fasciitis (fasciosis)

- Historically, thought was caused by inflammation.
- Currently, a degenerative rather than an inflammatory process.
- Characteristic symptoms of plantar fasciitis include:
  - An initial insidious onset of heel pain.
  - Intense heel pain during the first steps after waking or after a period of inactivity, with relief afterwards.
  - Pain that reduces with moderate activity but worsens later during the day or after long periods of standing or walking.
- Age, weight, and activities.



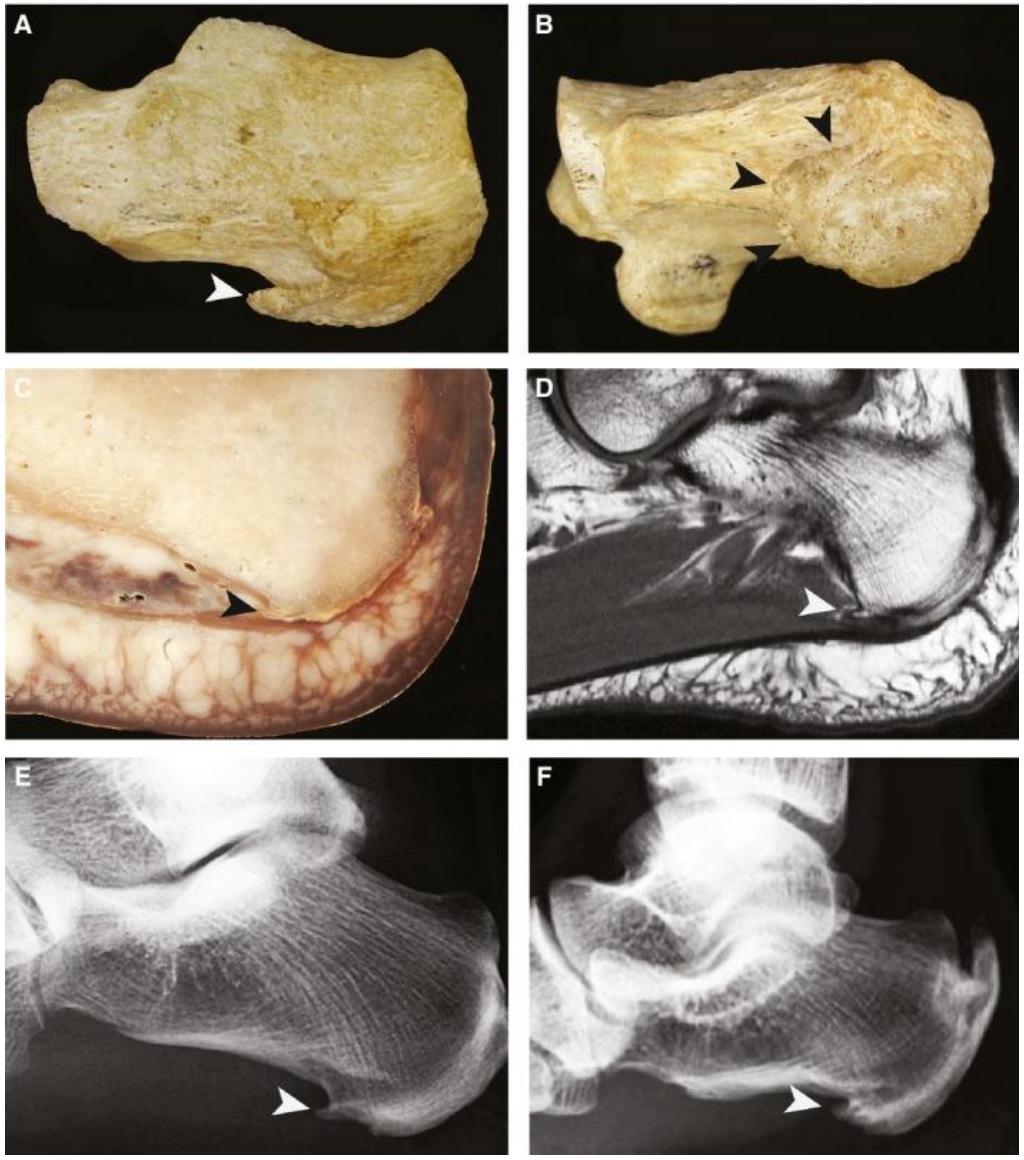
Plantar Fasciitis (fasciosis)



- Examine the foot at weight bearing and none weight bearing.
- Signs of plantar fasciitis include:
  - Tenderness on palpation of the plantar heel area.
  - Limited ankle dorsiflexion range (with the knee in extension).
  - Positive 'Windlass test'.
  - Tightness of the Achilles tendon.
  - An antalgic gait.
- If clinical features of plantar fasciitis are absent or inconsistent, consider an alternative diagnosis.

Calcaneal Spur

- Calcaneal spurs are fibro-cartilaginous triangular projections that vary in size present on the calcaneum.
- There are two types of calcaneal spur
- The dorsal calcaneal spur
- Plantar calcaneal spur .
- The etiology of these spurs is a contentious issue and it has been explained through a number of theories including the degenerative, inflammatory, traction, repetitive trauma, bone-formers and vertical compression theories.

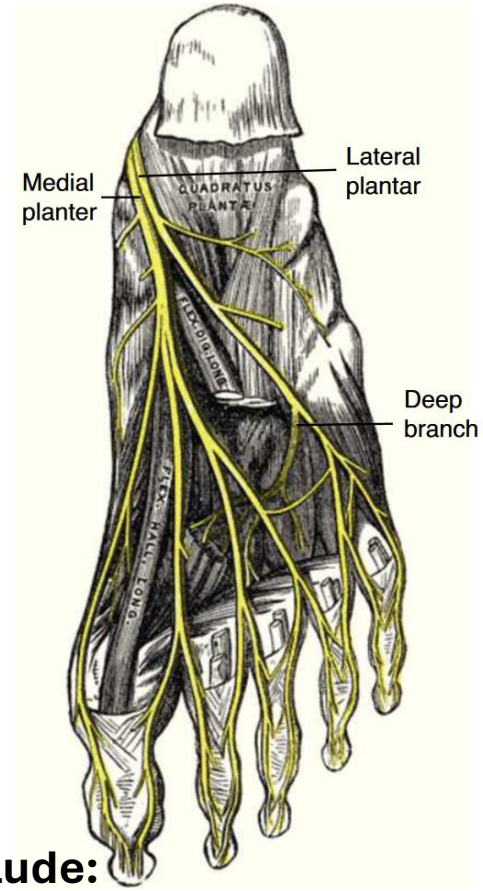


Calcaneal Spur

- A heel spur can be the result of longstanding tension on the plantar fascia insertion.
- However, most people who have bone spurs on their heels do not have heel pain.
- Plantar spurs, on the other hand, do not grow into the plantar fascia but their location is immediately deep to it.
- Since heel bone spurs are not the cause of plantar fasciitis, plantar fasciitis pain can be treated without removing the spur.



Morton’s neuroma



- Morton's neuroma is a compression neuropathy of the common digital plantar nerve.
- It occurs mostly in the third intermetatarsal space (66% of cases), less commonly in the second intermetatarsal space (30% of cases), and rarely in the first and fourth intermetatarsal spaces.
- A neuroma may rarely occur in more than one intermetatarsal space and is bilateral in 21% of cases.
- Morton’s neuroma is not a true neuroma, but a benign fibrotic thickening of the nerve due to constant irritation.

Mechanisms thought to cause irritation of the nerve include:

- Chronic, repetitive trauma.
- Nerve ischaemia.
- Intermetatarsal bursitis.
- Compression or entrapment of the nerve.

Risk factors include:

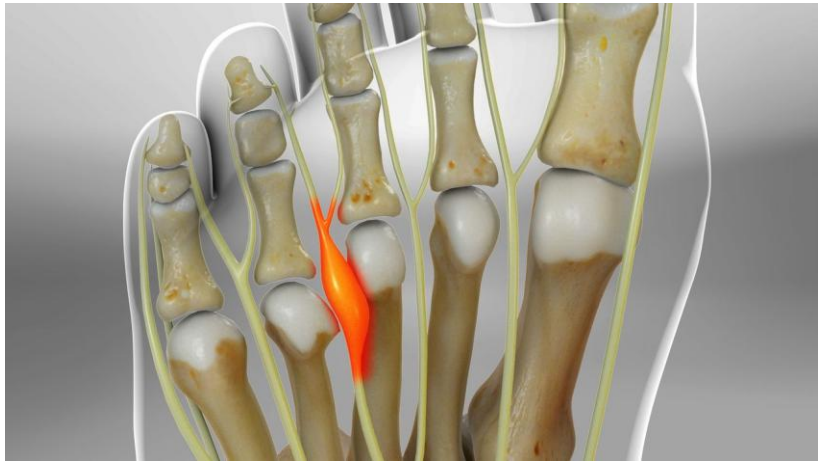
- High-arch feet.
- Regularly wearing tight/ill-fitting shoes.
- Repetitive, heavy impact on the feet, for example during running, dancing, or athletics.

Typical symptoms include:

- Pain in the forefoot, most commonly in the third intermetatarsal space.
- Pain whilst walking, exacerbated by increased activity or particular footwear, and relieved by removal of footwear and massaging the toes.
- The sensation of having a 'pebble' or 'lump' under the metatarsal region when walking.
- A sharp, stabbing, burning, or tingling sensation in the distribution of the affected nerve.

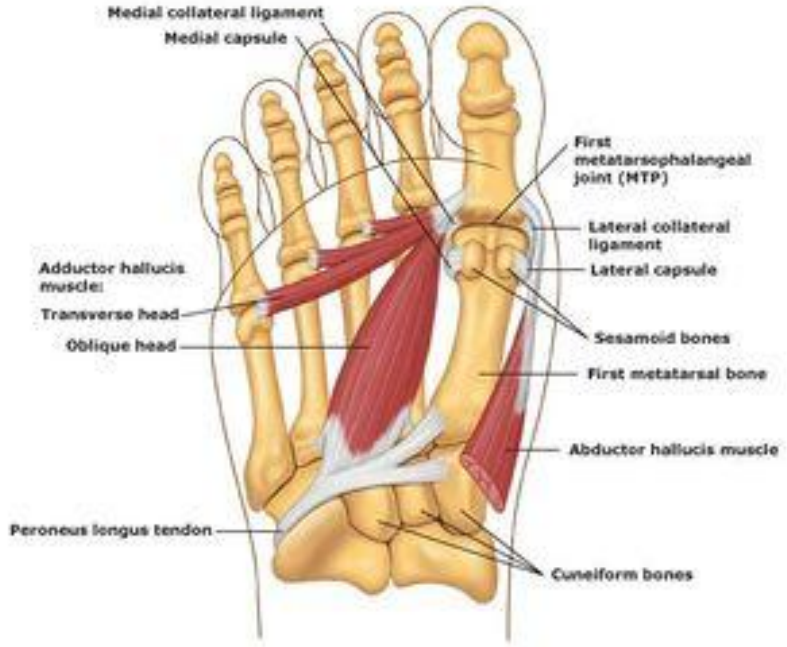
- 1st intermetatarsal space: Heuter neuroma
- 4th intermetatarsal space: Iselin neuroma

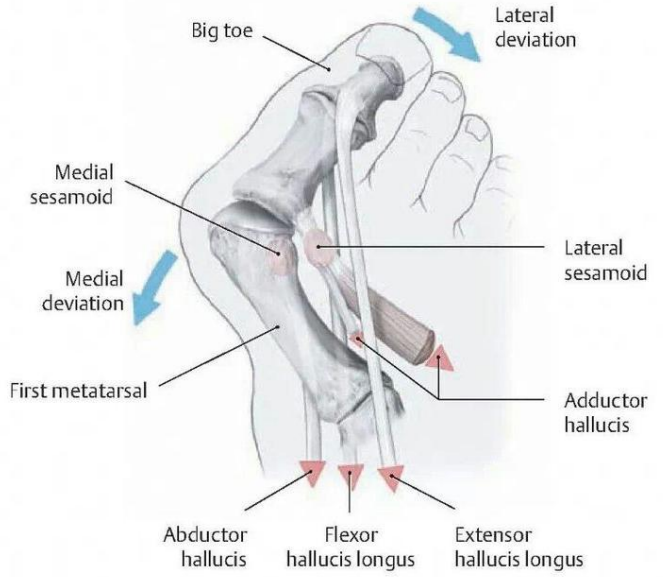




Hallux Valgus

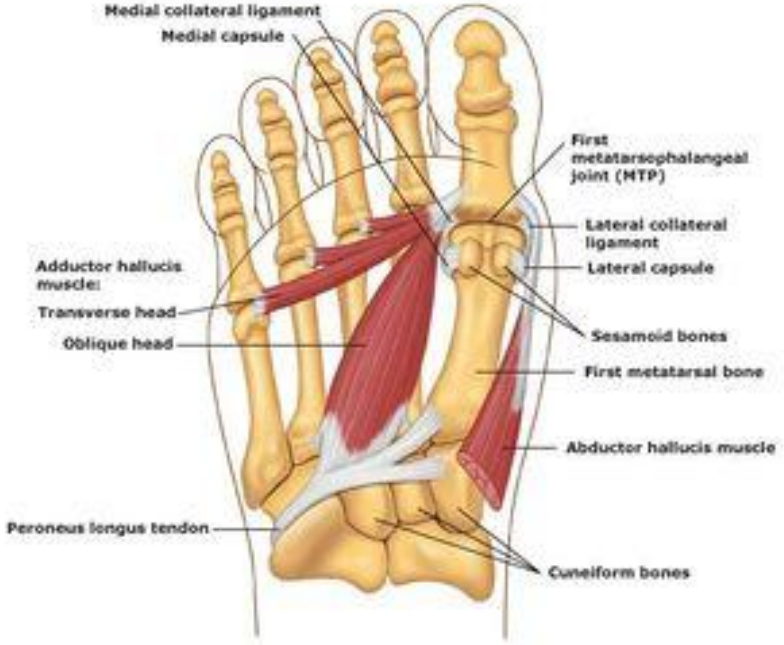
- The great toe (hallux) deviates laterally away from the midline towards the lesser toes.
- Less than 15° is considered normal. Angles of 20° and greater are considered abnormal. An angle of 45-50° is considered severe.
- This causes medial prominence of the first metatarsal head, and the overlying bursa and soft tissues may also become inflamed.
- Predisposing factors may include:
  - Poorly-fitting and constricting footwear.
  - age and female gender.
  - Family history.
  - Pes planus.
  - Achilles tendon contracture and ligamentous laxity.

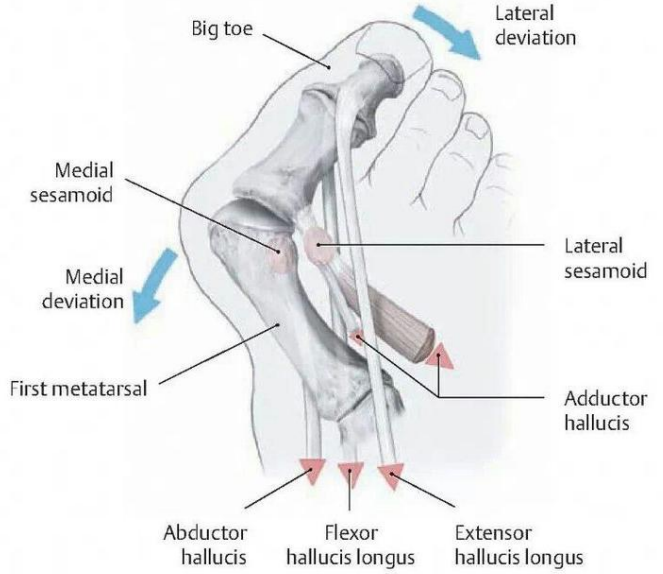




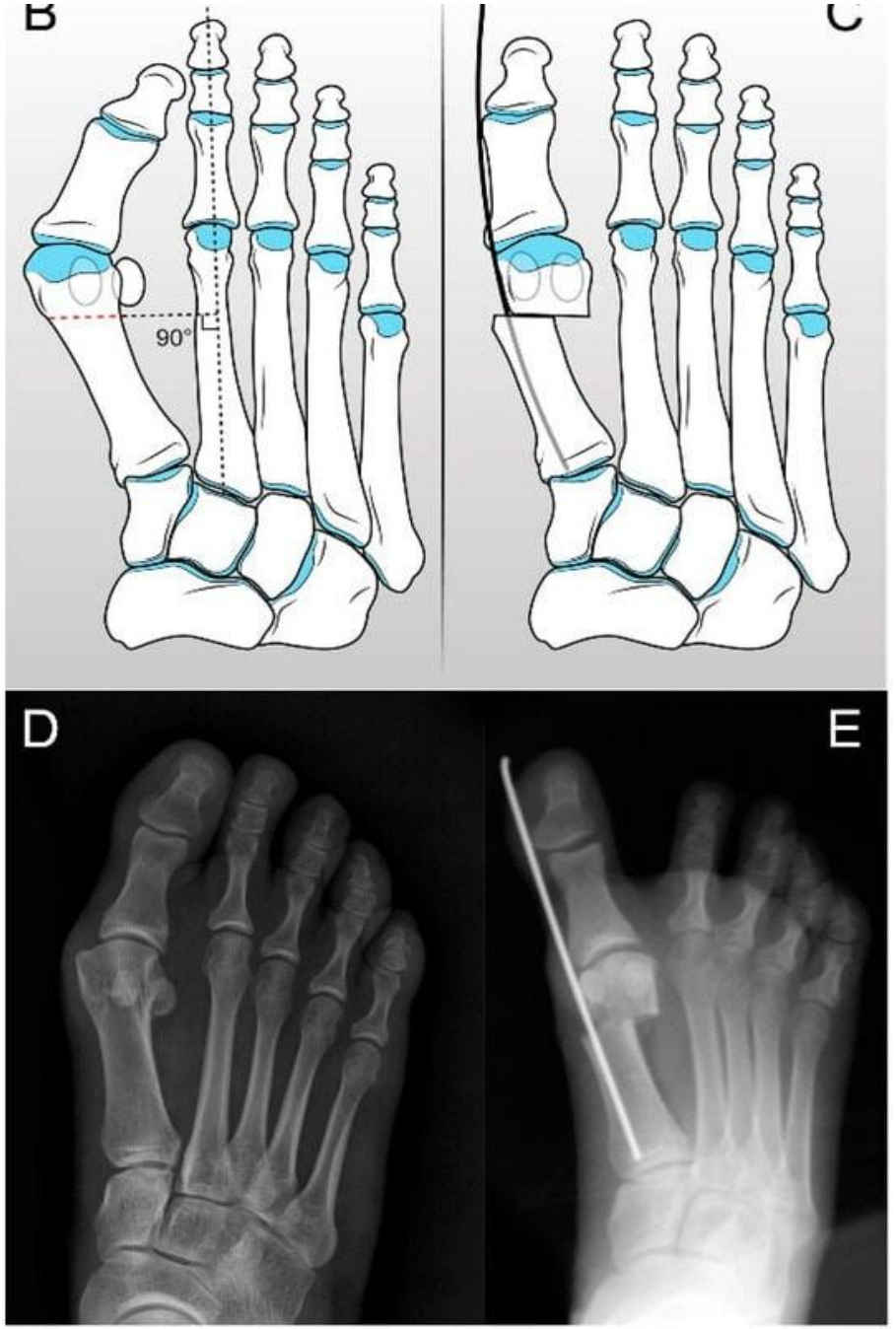
Hallux Valgus

- Initial injury relates to the failure of the medial sesamoid and the medial collateral ligaments
- Medial deviation of the first metatarsal head
- Valgus deformity of the proximal phalanx while its base is fixed to the sesamoid, transverse ligament and adductor hallucis tendon
- The lateral arching of the extensor and flexor hallucis longus tendons increases valgus deformity
- The adductor hallucis tendon pulls the phalanx in pronation, causing dislocation of the phalanx base
- First metatarsophalangeal joint rotates medially with pronation and becomes unstable



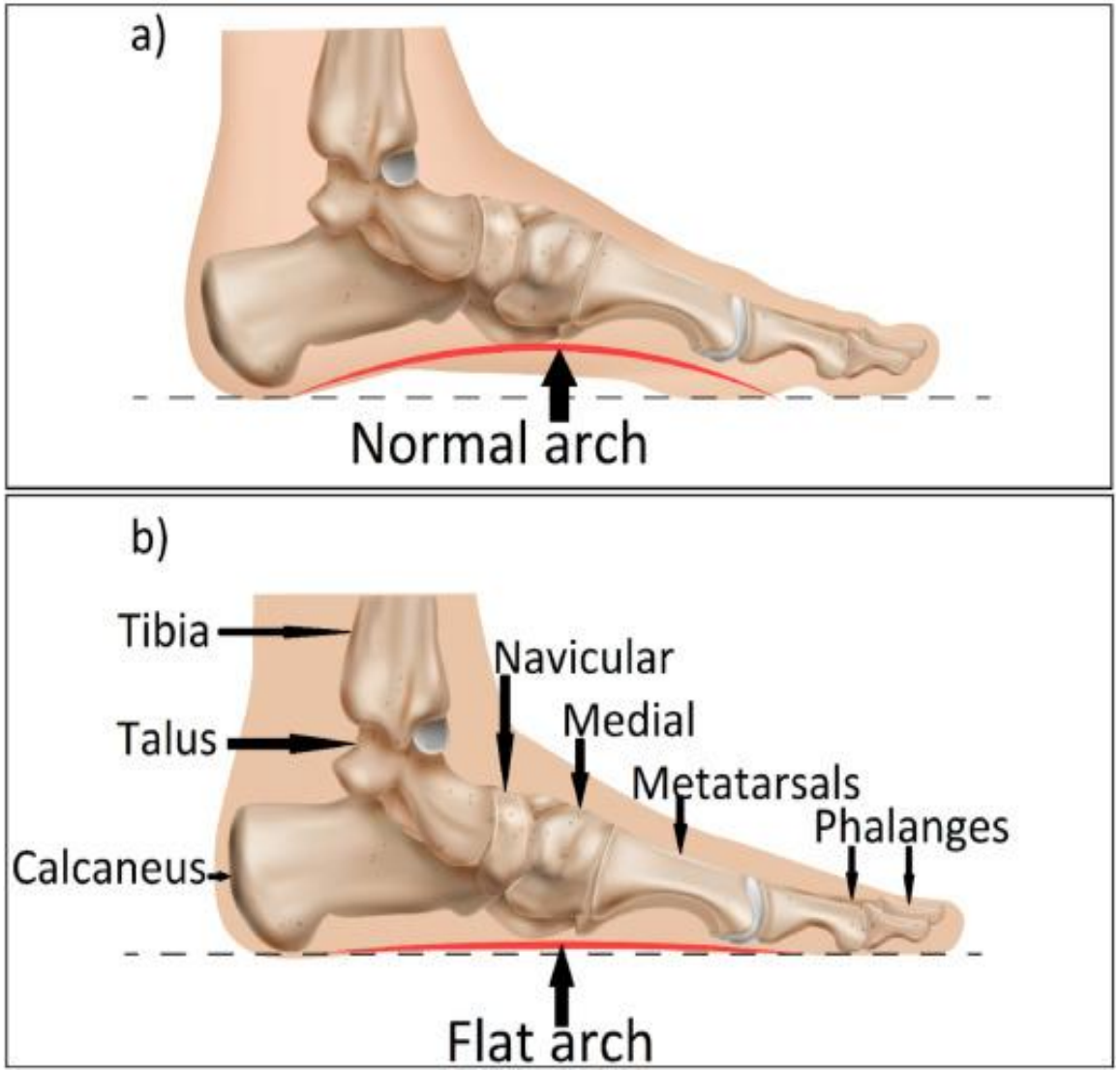


Hallux Valgus Surgery



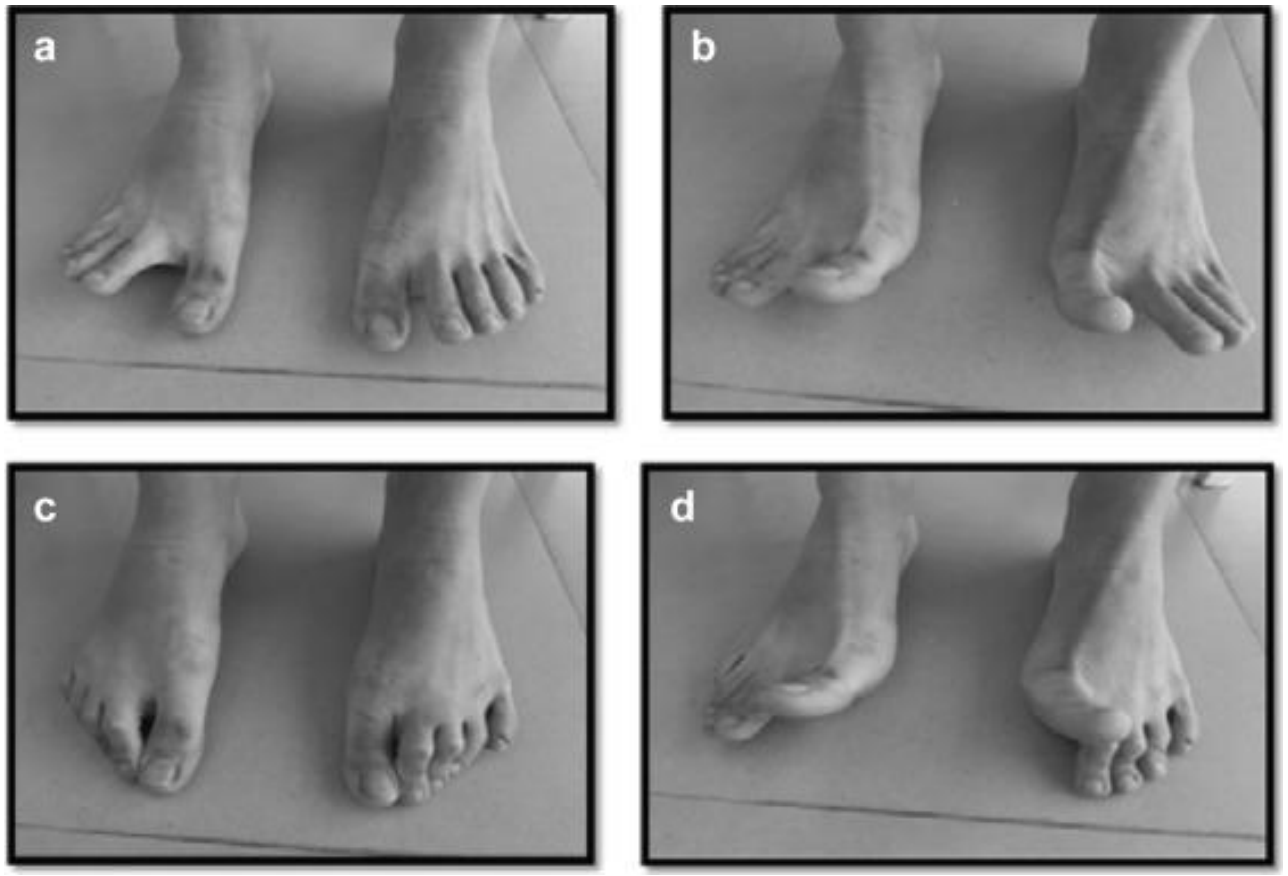
Pes planus

- There are several causes of flat feet, including genetics, injury, and underlying medical conditions.
- The congenital vertical talus that leads to a rigid flatfoot deformity is an example of a genetic factor.
- Acquired flatfoot can be linked to a range of factors, including a tight triceps surae or isolated gastrocnemius tightness, dysfunction of the posterior tibial tendon, midfoot laxity, abduction of the forefoot, external rotation of the hindfoot, subluxation of the talus, traumatic deformities, ruptured plantar fascia, Charcot's foot, and neuromuscular imbalance.
- Injuries to the foot, such as ligament and tendon damage, can also lead to the development of flat feet.



Exercises for PP.

- Dorsiflexion/plantarflexion,
- four short foot exercises: (toes extension, toes separation, big toe extension, foot doming)
- gluteal strengthening exercises and
- calf stretches.




Tibialis
Posterior
Strengthening
Interventions



Questions